EUROCONTROL

CNS Evolution Plan 2024

Laying the groundwork
for a comprehensive CNS strategy



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Executive summary

The CNS Evolution Plan will become the strategic reference to guide the modernisation and optimisation of the European Communications, Navigation and Surveillance (CNS) infrastructure, translating the vision established by the European ATM Master Plan 2025 Edition into an actionable roadmap.

This first version, Edition 2024, addresses Initiating Action 2 of the CNS Advisory Group Action Plan by defining the structure and content of the Evolution Plan, laying the groundwork for a more comprehensive strategy and outlining a structured methodology for its further refinement.

Future editions will build on this foundation to ensure that CNS systems evolve in a coordinated, efficient and forward-looking manner to meet the operational and strategic objectives of European and global airspace management.

In addition to defining a structure and methodology for the development of the Evolution Plan, the CNS Programme Manager has already developed three aspects of the current edition of the Evolution Plan and tested the first steps of the proposed methodology.

Firstly, a review of the vision and strategic drivers for CNS evolution was conducted, which highlighted the need to develop CNS systems that support the vision of the European ATM Master Plan, ICAO's Global Air Navigation Plan (GANP) and EASA's European Plan for Aviation Safety (EPAS) and ensure compliance with European and global regulatory frameworks.

In particular, it will be necessary to develop a complementary and detailed planning view to the high-level CNS roadmap in support of the ATM Master Plan Strategic Deployment Objectives #7 (Transition towards high performance of air-ground connectivity) and #9 (CNS optimisation, modernisation and resilience).

Secondly, the CNS Programme Manager initiated a survey of its stakeholders to verify the existence of issues with the current CNS infrastructure and its evolution, and the need to address them through the development of the CNS Evolution Plan to achieve the aforementioned vision and strategic objectives. This issue draws the attention of the stakeholder community to short term issues identified by the CNS Programme Manager as part of his work.

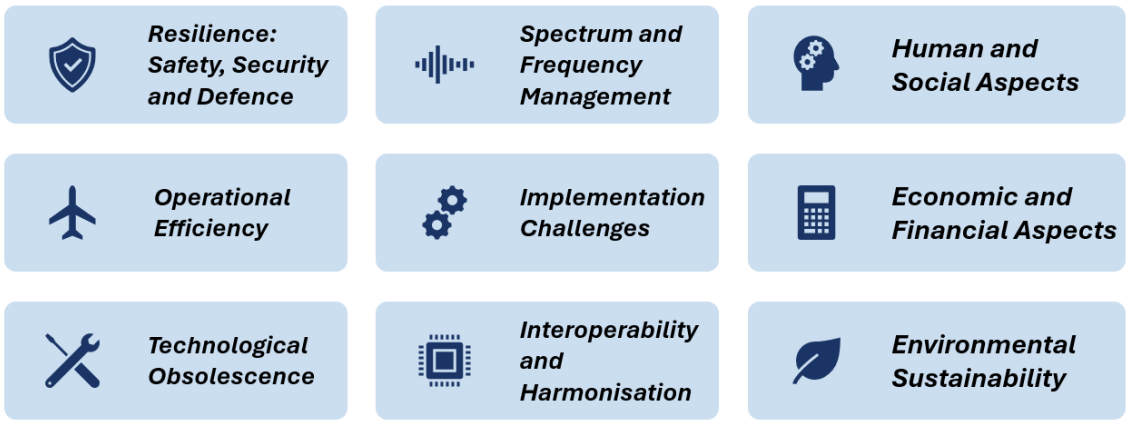


Figure 1 - Key objectives of CNS Evolution

Key CNS Challenges for 2025 include:

- ***Implementing EASA SIB No. 2022-02R3 recommendations to address GNSS outage and alterations leading to CNS degradation;***
- ***Defining local CNS infrastructure plans considering the European MON principles & design criteria;***
- ***Evaluating intervention strategies in relation to PBN IR ILS CAT I;***
- ***Addressing current Datalink performance issues and take steps towards multilink to enable future applications;***
- ***Addressing over-interrogation in 1030 MHz band; and***
- ***Contributing to an EU Aviation Spectrum Strategy.***

These challenges should be addressed by the relevant actors as soon as possible, as they can have a significant impact on the CNS modernisation and the performance of the network.

Thirdly and lastly, the CNS Programme Manager initiated a survey to identify current and emerging CNS technologies that should be considered in the development of deployment scenarios, their screening, and their assessment against the objectives outlined in the problem analysis. This edition of the Evolution Plan contains an abstract of this comprehensive inventory, which is maintained on a bespoke interactive dashboard.

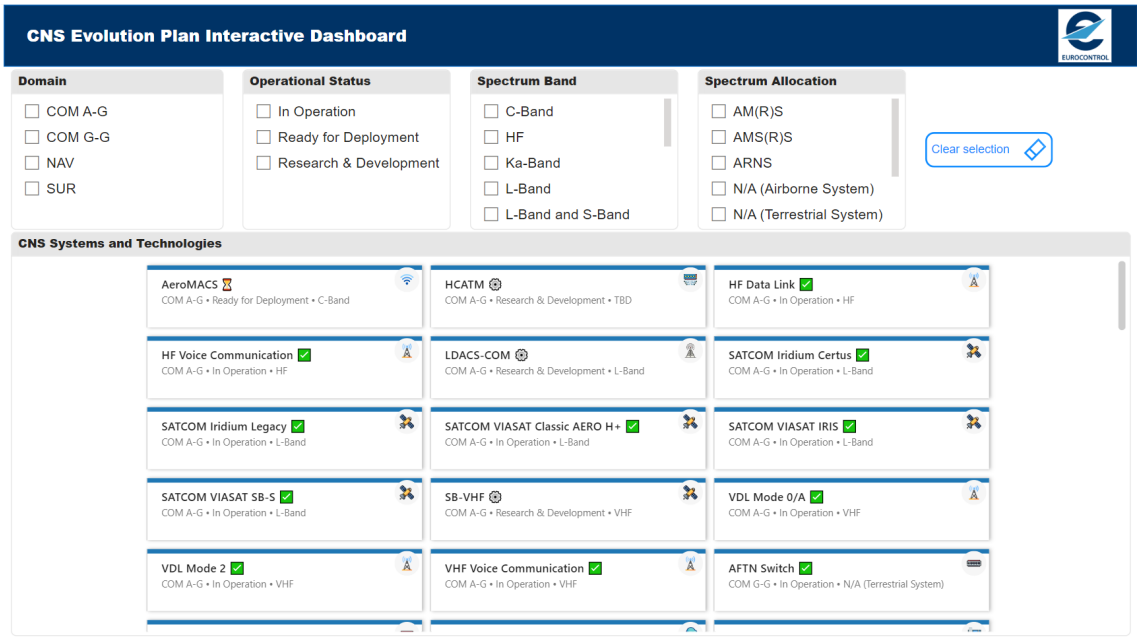


Figure 2 - CNS Evolution Plan Interactive Dashboard (illustrative content)

The 2026 edition of the CNS Evolution Plan will include a more comprehensive and in-depth analysis of the underlying causes of the issues, as well as a refined assessment of the options; supported as far as reasonably practicable by tangible and quantified evidence and using the approved methodology.

The plan will encompass all Member States within the Single European Sky (SES) as well as the broader European Civil Aviation Conference (ECAC) Area. The long-term perspective will extend up to 2050, aligning with technological advancements and evolving operational requirements to ensure that CNS systems can meet the increasing demand for air traffic management, sustainability, and global competitiveness.

The plan will also consolidate the CNS modernisation aspects with the optimisation of legacy CNS infrastructure, taking into account the national MONs designed using the

common design principles but based on local requirements and constraints, and captured using established implementation reporting mechanisms.

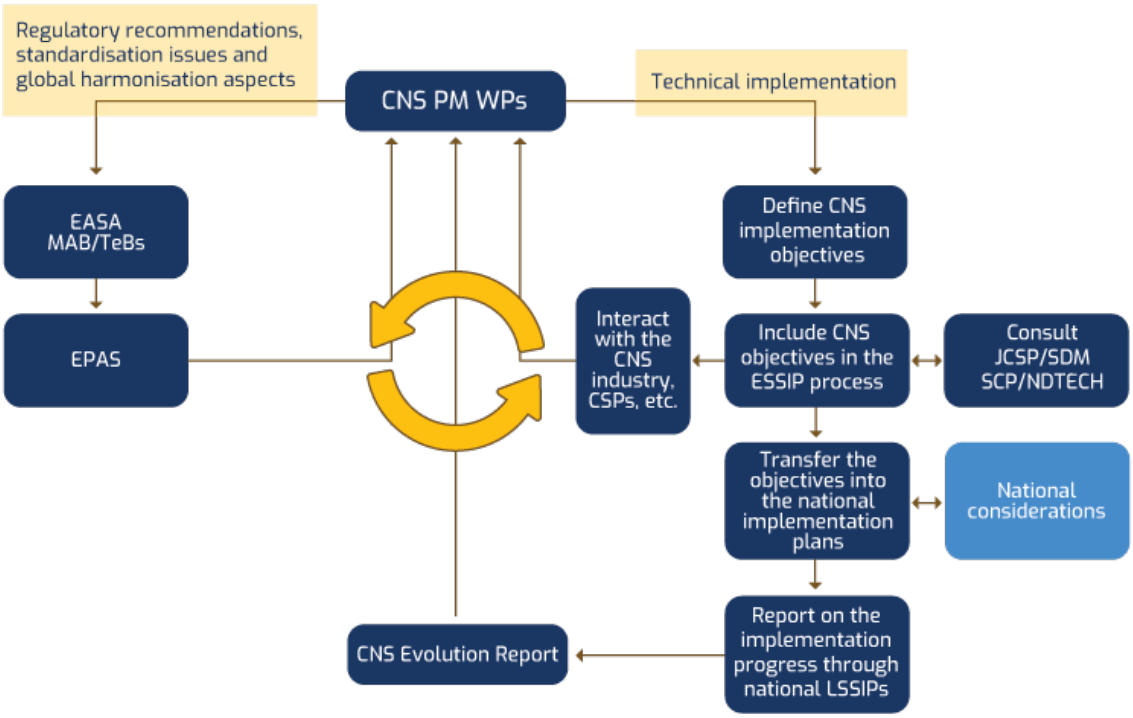


Figure 3 - Proposed CNS Evolution Plan monitoring and reporting process

1 Introduction

1.1 CNS Advisory Group Report

(1) The CNS Advisory Group was established by DG MOVE and started to work in 2020 to develop recommendations for a better organisation and management of the CNS infrastructure, noting the well-documented issues associated with the implementation of past EU CNS mandates.

(2) The CNS Advisory Group Report [1] was reviewed by Operational stakeholders, and representatives of the manufacturing industry and other stakeholders.

(3) Key issues Identified by the CNS Advisory Group:

a) Pan-European CNS deployment is slow.

b) The CNS infrastructure redundancy layers and geographic locations are not optimised.

c) The variety of airspace users' needs is an important consideration for the efficient deployment.

d) CNS infrastructure is not sufficiently spectrum-efficient and is not sufficiently protected against physical and cyber security threats.

e) CNS interoperability requirements and compliance demonstration have not been fully addressed. The lack of a harmonised compliance framework resulted in additional delays, duplication of efforts, and costs for airspace users and ANSPs.

f) There is a lack of European wide CNS Programme Management addressing the full life cycle of the CNS infrastructure.

(4) Report findings:

a) The CNS advisory group is fully aware of the fact that the CNS infrastructure is not delivering operational benefits directly; it is a technical enabler to provide operational benefits to ATM and airspace users. The evolution of the CNS infrastructure, including both deployment of new CNS systems and optimisation of some legacy CNS systems, is needed to achieve the long-term objectives of ATM modernisation in support of the evolution of Air Traffic as defined in the European ATM Master Plan.

b) The current CNS infrastructure provides a robust and resilient service. However, occasionally CNS outages can cause a significant degradation of ATM services.

c) The targeted CNS infrastructure will be composed of an optimised combination of ground and space-based technologies, complemented with Minimal Operational Networks (MONs) of conventional infrastructure that are needed to maintain services in case outages occur in the primary infrastructure (e.g. interferences affecting satellite navigation signals).

(5) Overall, the Advisory Group made a series of recommendations, requiring buy-in from the complete stakeholder community if they were to be met. One of these recommendations was to '' *Translate the Master Plan's CNS roadmap into a 'CNS evolution plan' with short, medium and long-term objectives, priorities and decision points.*" These recommendations were accompanied by an 18-point "Action Plan for the development and management of the implementation of the European

CNS evolution plan." It was broken down into three Initiation actions, a further three Main actions and twelve Supporting actions and processes.

(6) The EUROCONTROL delegation to the CNS advisory group volunteered to produce also an economic assessment [2] based on data available to the Network Manager; the proposal was accepted by the Chair. The study was titled the "Economic perspective" and was enshrined in Chapter 5 of said report.

(7) This document is a product of the CNS Programme Manager, a task carried out by EUROCONTROL in its role as Network Manager (NM) in full partnership with all relevant stakeholders.

(8) NM was given the task of The CNS Programme Manager based on a recommendation from the CNS Advisory Group and in application of Article 7 of the Regulation (EU) 2019/123 (NF IR), to manage the implementation of a European CNS Evolution Plan, comprising both infrastructure optimisation and modernisation elements.

1.2 Document Purpose

(1) The CNS Evolution Plan aims to provide a comprehensive CNS optimisation and evolution approach for the European stakeholders, with clear milestones for key decisions, and building on the joint efforts of civil and military stakeholders to ensure synergies, while keeping realistic expectations. CNS Evolution Plan is fully consistent with the European ATM Master Plan that provides the vision for ATM evolution in Europe. The evolution and operation of the CNS infrastructure should also directly contribute to the decarbonisation of aviation in line with the EU Green Deal objectives and facilitate the integration of new entrants and technologies. The solutions and technologies selected in the CNS Evolution Plan should be suitable to support operational improvements in all anticipated operational scenarios

(2) This initial version of the CNS Evolution Plan is a preliminary version and serves as a foundational document outlining the key objectives of CNS evolution in **Section 3**. It highlights the current challenges and sets the groundwork for the future, more comprehensive CNS Evolution Plan development.

(3) The CNS Evolution Plan is consistent with key European and global strategic documents such as the European ATM Master Plan, ICAO's Global Air Navigation Plan (GANP), EASA's European Plan for Aviation Safety (EPAS), the Network Strategy Plan, and the European Radio Navigation Plan. These plans serve as the strategic drivers outlined in **Section 2** of this document. The complete CNS Evolution Plan will not only align with the aforementioned strategic frameworks but will also actively contribute to their evolution by identifying necessary changes and proposing improvements.,

(4) In line with the overall planning outlined in the CNS Advisory Group Action Plan (Figure 4) [3], this edition of the CNS Evolution Plan relates to initiation action 2, in particular the important aspects of definition of the structure and content of the CNS Evolution Plan, including a methodology for its development and maintenance included in **Appendix 1**.

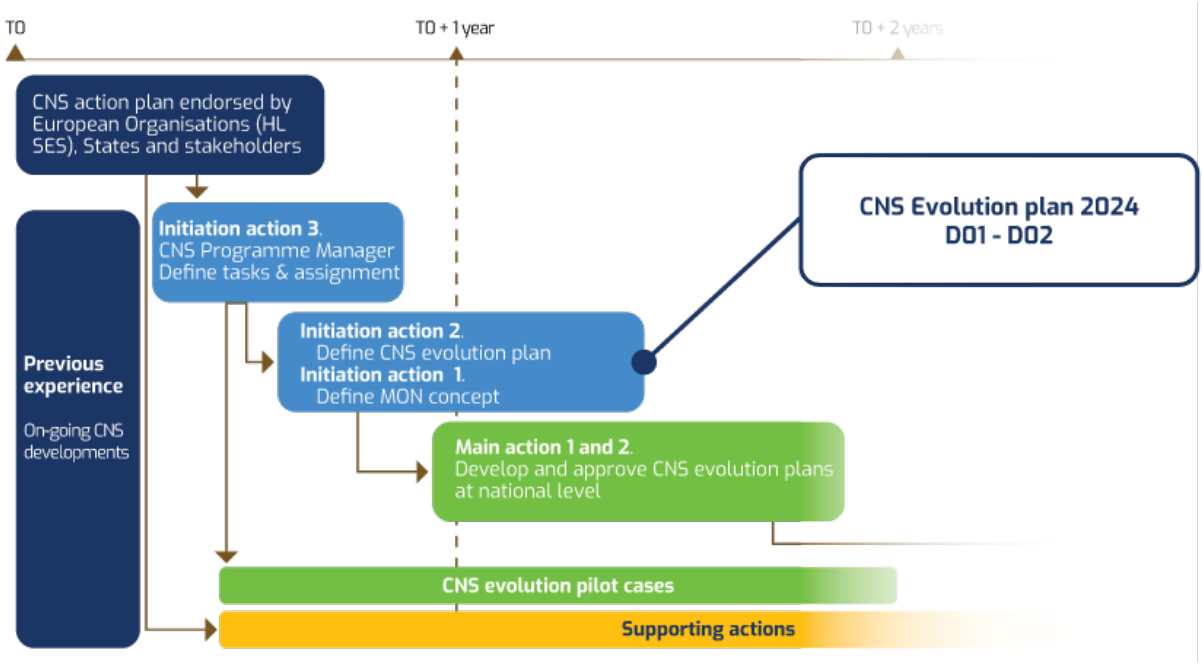


Figure 4 - CNS Advisory Group Action Plan

(5) The CNS Programme Manager will further develop the CNS Evolution Plan, considering national MONs, and will eventually deliver a consolidated Evolution Plan including both optimisation and deployment elements, fulfilling main actions from the Action Plan (details provided in **Section 6**). This is planned for the 2026 edition of the CNS Evolution Plan, which is expected to be the first one to fully apply the development methodology.

1.3 Geographical Scope

(1) Two geographical scopes are considered and evaluated throughout this document. First, the Member States (MS) subject to the Single European Sky[1]. Second, the European Civil Aviation Conference (ECAC) Area given the Pan-European nature of network management.

(2) In both cases, the main focus of the analysis will be on continental airspace, which is over land and has more CNS infrastructure However, the topics and issues discussed in this document will include oceanic and remote airspaces, ensuring a comprehensive consideration of the full airspace environment, including those areas with less dense infrastructure.

1.4 Time Horizon

(1) The CNS Evolution Plan is designed with a long-term perspective, covering the timeframe up to 2050. This horizon reflects the evolving nature of aviation technology, airspace management needs, and the increasing demand for sustainability and operational efficiency.

(2) The period leading to 2050 will witness significant changes in aviation, with the introduction of new technologies, airspace users, and operational concepts. The CNS Evolution Plan outlines a phased approach, beginning with short-term goals

[1] 'Single European Sky airspace' means airspace above the territory to which the Treaties apply, as well as any other airspace where Member States apply Regulation (EC) No 551/2004 in accordance with Article 1(3) of that Regulation.

aimed at addressing immediate regulatory, technical, and operational challenges, followed by medium-term milestones that focus on widespread adoption of advanced CNS systems, and long-term strategies for achieving full integration of emerging technologies, such as space-based CNS systems, Innovative Air Mobility (IAM) and Drones operations.

(3) By 2050, the aviation industry is expected to face unprecedented demand for air traffic management, requiring robust CNS systems that can support increased traffic, ensure environmental sustainability, and maintain the highest standards of safety and security. The CNS Evolution Plan will provide a roadmap for the continuous modernisation of CNS infrastructure, ensuring that Europe's aviation sector remains globally competitive and aligned with international standards.

(4) The CNS Evolution Plan will be adaptable to future developments in technology and operations, fostering a forward-thinking approach to CNS modernisation. It will serve as a living document, periodically updated to reflect new innovations, regulatory changes, and emerging operational requirements, ensuring that the CNS infrastructure remains resilient and capable of supporting the future of aviation.

1.5 Intended Audience

(1) Primarily this document is relevant to Member States, which are responsible for establishing the national operational requirements and spectrum protection; civil and military Air Navigation Service Providers (ANSPs) and aerodrome operators, which own and operate the CNS infrastructure; aircraft operators, which rely on CNS services for safe and efficient operations. Additionally, it is relevant to National (civil and military) Competent Authorities, which are responsible for certification of ANSPs and monitoring their compliance with applicable regulations.

(2) The European Union Aviation Safety Agency (EASA), which is responsible for ensuring the highest common level of safety and environmental protection for EU citizens through the development and adoption of common civil aviation safety rules, and measures, provides input into CNS Evolution Plan. This evolution planning is done in close coordination with the European Union Aviation Safety Agency (EASA) to ensure alignment between the Plan and the regulatory framework to accelerate deployment. EASA also acts as the competent authority for certain ATM/ANS and U-space service providers, particularly all pan-European service providers.

(3) In addition, this document is also relevant for institutional partners that have a responsibilities in relation to CNS, including ICAO, the European Commission, SESAR Deployment Manager (SDM), the SESAR3 Joint Undertaking (S3JU), the Performance Review Body (PRB), the European Defence Agency (EDA), NATO, European Space Agency (ESA), European Union Agency for the Space Programme (EUSPA), as well as for the CNS manufacturing industry, Communications Service Providers (CSPs), aircraft and equipment manufacturers, airspace users, R&D organisations, professional staff organisations, relevant trade associations and standardisation organisations.

1.6 Document Structure

(1) The current version of the CNS Evolution Plan will be structured as follows:

a) **Section 1** introduces the purpose, geographical scope, and intended audience of the CNS Evolution Plan, setting the context for its development and alignment with broader aviation strategic documents.

b) **Section 2** discusses the strategic frameworks driving the CNS evolution, including the European ATM Master Plan, ICAO GANP, EASA EPAS, and other key documents. It outlines the overarching vision and objectives for modernising CNS systems in Europe.

c) **Section 3** identifies the main challenges faced by CNS infrastructure and defines the initial objectives of the CNS Evolution Plan, including interoperability, spectrum management, safety, and environmental sustainability.

d) **Section 4** gives an overview of current and emerging CNS technologies, focusing on their operational status and future development. This section covers the domains of Communications (COM), Navigation (NAV), and Surveillance (SUR), along with systems currently in operation and those under research and development.

e) **Section 5** reserved as a placeholder for the content of the 2026 edition of the CNS Evolution Plan.

f) **Section 6** is a concluding section describing the Implementation and Monitoring process for the CNS Evolution Plan.

g) **Appendices** provide additional supporting information, including development methodology, regulatory frameworks, and the European ATM Master Plan cross-references.

2 Vision and Strategic drivers of CNS Evolution

(1) The development of the CNS Evolution Plan will address the operational needs and requirements of ATM in Europe and globally, ensuring the modernisation of CNS infrastructure in alignment with the evolving demands of airspace operations. This modernisation is critical for achieving the long-term goals of environmental sustainability, operational efficiency, and seamless coordination across all domains of air traffic management.

(2) The CNS Evolution Plan will support the development of a modern, interoperable, and efficient digital European sky, driven by the vision set forth in the European ATM Master Plan and aligned with ICAO's Global Air Navigation Plan (GANP), European Plan for Aviation Safety (EPAS), Network Strategy Plan, and European Radio Navigation Plan.

(3) CNS Evolution Plan tends to optimise CNS infrastructure in Europe and, where reasonable, to optimise infrastructure where it does not negatively affect aviation safety, ATM capacity nor national security, taking into account global security situation which is relevant to EU countries.

(4) As the first edition of the CNS Evolution Plan, it aims to establish a foundation for alignment with the broader goals of global and regional air traffic management systems. The full edition of CNS Evolution Plan (2026) will be designed to be responsive to the evolving needs of the aviation industry, addressing key challenges.

2.1 European ATM Master Plan

(1) The CNS Evolution Plan is fully aligned with the ATM Master Plan's vision [4] of making Europe the most efficient and environmentally friendly sky to fly in the world by 2045. By addressing the detailed requirements for CNS infrastructure, it will enhance the roadmap's ability to meet the dynamic needs of emerging applications, innovative air mobility (IAM), and increasingly complex airspace operations. In this context, the CNS Evolution Plan will act as a bridge, ensuring the operationalisation of the high-level strategies within the ATM Master Plan and facilitating the coordinated deployment of CNS solutions across Europe.

(2) The CNS Evolution Plan will serve as a complementary and detailed planning view to the high-level CNS roadmap outlined in the European ATM Master Plan 2025 Edition. The Master Plan, particularly through its focus on Strategic Deployment Objectives (SDO) shown on Figure 5, such as SDO7 on transition towards high performance of air-ground connectivity (multilink) and SDO9 on CNS optimisation, modernisation, and resilience, already establishes the strategic direction for CNS improvements.

(3) **SDO 7 – Transition towards high performance of air-ground connectivity (multilink)**

(4) This strategic deployment objective aims to improve air-ground datalink performance, accelerate technology adoption, and achieve economies of scale while reducing the fragmentation that characterises current datalink implementation.

(5) **SDO 9 - CNS optimisation, modernisation and resilience**

(6) This strategic deployment objective aims to optimise, modernise, and increase the resilience of CNS infrastructure in Europe.



Figure 5 - European ATM Master Plan 2025 Strategic Deployment Objectives

(7) The CNS Evolution Plan will be built upon this foundation by providing specific technical and operational details necessary to implement the high-level strategies outlined in the Master Plan. It will delve deeper into the technical specifications, timelines, and operational frameworks required to achieve the envisioned transformations, particularly regarding CNS technologies. More detailed information about consistency between CNS Evolution Plan 2024 and the European ATM Master Plan 2025 is presented in **Appendix 3.**

2.2 ICAO Global Air Navigation Plan

(1) The ICAO Global Air Navigation Plan (GANP) serves as a critical framework for defining key Air Traffic Management (ATM) operational requirements, driving global harmonisation and modernisation of airspace operations. The GANP, particularly through the Aviation System Block Upgrades (ASBU) framework (Figure 6), provides a strategic roadmap for the phased deployment of CNS technologies.

(2) The following CNS elements are represented in the current 6th Edition of the GANP [5]:

(3) **Surveillance Systems (ASUR)** with the aim of modernising and enhancing surveillance capabilities, particularly in remote and oceanic regions.

a) Current (Block 0): Use of ADS-B OUT and Multilateration (MLAT) systems, complementing traditional radar.

b) Future (Blocks 1-3): ADS-B surveillance via space-based systems and the evolution of vehicle-to-vehicle surveillance for autonomous operations. By Block 3, performance-based surveillance will allow for flexible use of surveillance technologies tailored to specific airspace needs.

(4) **Communications Infrastructure (COMI)** focused on improving and expanding communications infrastructure, integrating new technologies to support higher capacity and efficiency.

a) Current (Block 0): Use of VHF, HF, and SATCOM for voice communications, with increasing reliance on data links like VDL Mode 2 and SATCOM Class C for messaging.

b) Future (Blocks 1-3): Introduction of multi-frequency VDL Mode 2 and ATN/IPS for data exchange, eventually leading to broadband communications and SATCOM Class A systems for robust, global coverage and increased capacity.

(5) **ATS Communications Service (COMS)** aiming to move from voice-based to data link communications, improving air-ground communications and reducing congestion.

a) Current (Block 0): Introduction of Controller-Pilot Data Link Communications (CPDLC) for reducing voice channel congestion, particularly in procedural airspace.

b) Future (Blocks 1-3): Full extension of CPDLC and ADS-C systems, leading to increased automation of aircraft systems and trajectory synchronisation, supporting dynamic and flexible airspace operations.

(6) **Navigation Systems (NAVS) designed** to advance satellite-based and ground-based navigation systems, ensuring precision and resilience in all phases of flight.

a) Current (Block 0): Use of GNSS with augmentations like SBAS and GBAS, and introduction of Minimal Operating Networks for optimisation of conventional navigation aid infrastructure.

b) Future (Blocks 1-4): Implementation of Dual-Frequency Multi-Constellation (DF/MC) GNSS for greater robustness. **Alternative PNT (A-PNT)** solutions will be developed to act as complementary means to GNSS during disruptions, ensuring navigation continuity.

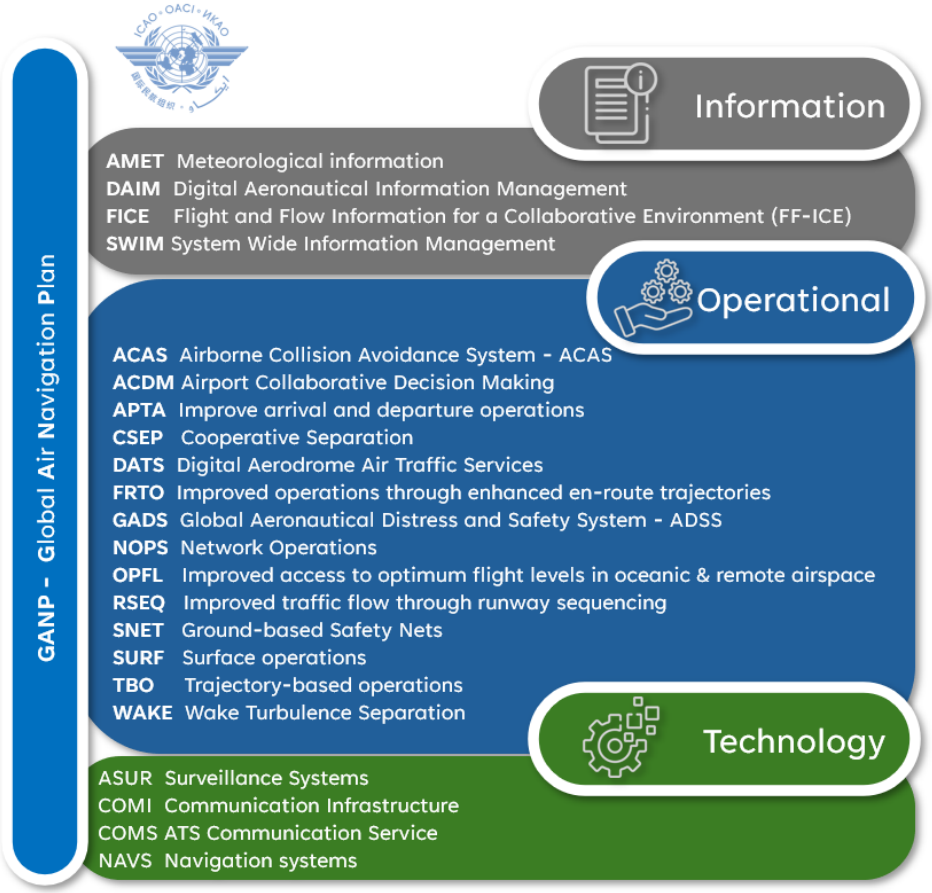


Figure 6 - ICAO GANP ASBU Threads

(7) In parallel, ICAO is actively developing CNS and Spectrum roadmaps through the Integrated CNS and Spectrum Task Force (ICNSS-TF), which will offer strategic guidance for the transition to more integrated and efficient CNS systems. The ICNSS-TF CNS Technology Roadmap, currently in progress, is planned for publication in 2025. At the same time, the ICAO GANP Study Group is working on the 7th Edition of the ICAO GANP, which is also set for release in 2025. These two roadmaps will complement each other: the updated GANP will focus on near-term operational improvements, while the ICNSS-TF roadmap will provide a more comprehensive, long-term vision for CNS systems development.

2.3 European Plan for Aviation Safety

(1) The CNS Evolution Plan will be closely aligned with the safety priorities set out in the European Plan for Aviation Safety (EPAS) and also be aligned with the related regulatory activities included in the EPAS. The EPAS [6] outlines the strategic safety priorities (Figure 7) for the European aviation system, aiming to maintain and enhance aviation safety while also supporting environmental sustainability and efficiency. The CNS Evolution Plan will serve as a complementary framework, supporting the safety objectives of EPAS by providing a detailed roadmap for the modernisation and integration of CNS systems.

(2) A key aspect of the CNS Evolution Plan is ensuring that the development and implementation of new CNS technologies contribute to the safety goals outlined in EPAS. These goals include fostering systemic safety and resilience, managing risk interdependencies, and improving safety management. The CNS Evolution

Plan will focus on advancing CNS infrastructure to support these objectives, promoting the seamless integration of modernised systems while maintaining high safety standards across the European aviation sector.

(3) Key EPAS priorities, such as managing interdependent risks, enhancing cybersecurity in order to improve safety, and incorporating new aviation domains like unmanned aircraft systems, will be supported through CNS advancements detailed in the Evolution Plan. Furthermore, as EPAS targets ensuring a level playing field and fostering a sustainable aviation system, the CNS Evolution Plan will provide a technical roadmap that enables these objectives through upgraded CNS infrastructure and services.

(4) In line with the EPAS's emphasis on decarbonisation and sustainability, the CNS Evolution Plan will also contribute to the environmental objectives set out in the European Green Deal and initiatives like "Destination 2050." The modernisation of CNS systems will be critical in enabling more efficient flight operations, optimising trajectories, and reducing the overall environmental impact of aviation, all while maintaining high safety standards. Through this alignment, the CNS Evolution Plan will play a crucial role in supporting both the safety and environmental priorities established in EPAS.

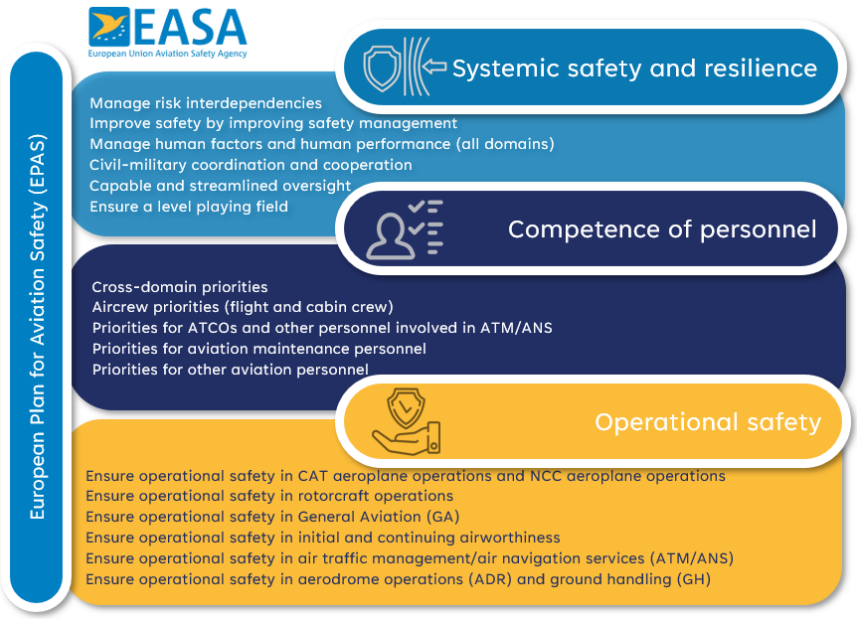


Figure 7 - EASA EPAS Strategic Priorities

2.4 Network Strategy Plan

(1) The EUROCONTROL Network Strategy Plan (NSP) for 2025-2029 sets out the main steps needed to support the achievement of the Network Performance for the next Reference Period of the SES and to support the future evolutions. The NSP is aligned with ICAO work in the field of ATM and the European ATM Master Plan. The vision of the NSP [7] is to achieve a European ATM network serving European civil and military aviation and passengers in a safe, secure, predictable, operationally efficient, environmentally friendly and cost-efficient manner through close cooperation with all operational stakeholders and includes Network Strategic Objectives (Figure 8). A key Strategic Objective particularly relevant to the ongoing development of the CNS Evolution Plan is **Strategic Objective 7 (SO 7): "Optimise and Digitalise (as appropriate) ATM/CNS**

infrastructure and services to support evolutions towards more efficient network operations and services".

(2) The NSP recognises that the main strategic step for this Strategic Objective will be the full implementation of the CNS Programme Manager Concept that will integrate CNS infrastructure planning, optimisation and monitoring into a single value chain approach. This Strategic Objective will drive evolution towards the Network Infrastructure Performance and Cost Effectiveness, through the implementation of a digital open architecture. The European aviation infrastructure will then become more sustainably performant, cost effective, and support the integration of new entrants.

(3) As the CNS Evolution Plan is designed to enhance CNS systems across Europe, it plays a crucial role in achieving the objectives set out in SO 7. Specifically, the CNS Evolution Plan will provide detailed guidance and frameworks to support the optimisation and modernisation of CNS systems, which is a key priority under SO 7. In this context, the CNS Evolution Plan will also serve as a foundation for meeting the broader network objectives outlined in the NSP, such as improving cross-border interoperability, enhancing ATM efficiency, and facilitating the integration of new entrants and technologies.

(4) Both plans emphasise sustainability and digital transformation by encouraging the use of open architectures and digital solutions, which aligns with the NSP's vision for the future of European airspace operations. The CNS Evolution Plan's ongoing development will ensure that it remains aligned with the evolving strategic direction set forth in the NSP, ensuring the efficient and secure transformation of Europe's CNS infrastructure.

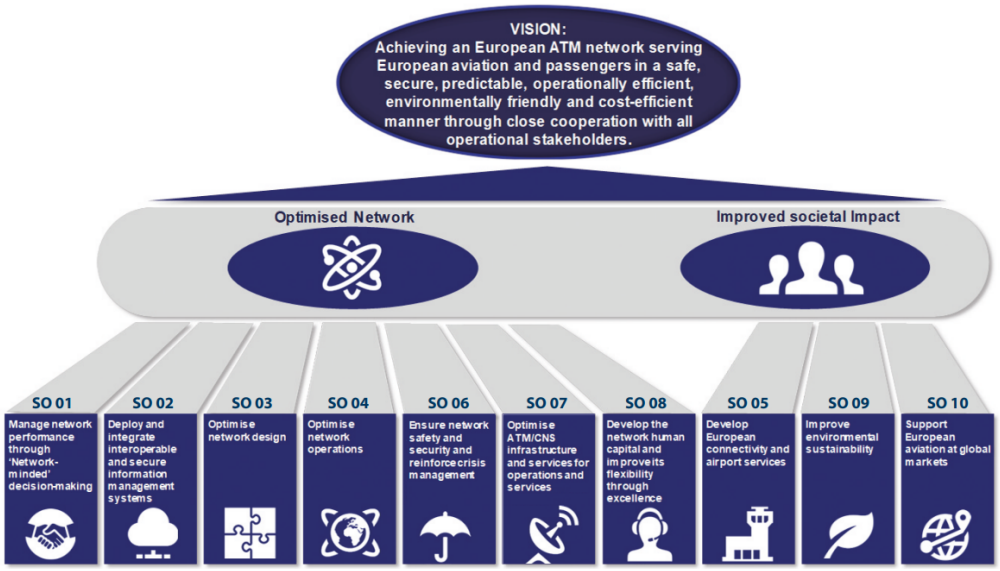


Figure 8 - NSP Vision and Strategic Objectives

2.5 European Radio Navigation Plan

(1) The European Radio Navigation Plan (ERNP) [8], developed by the European Commission, outlines the strategic vision for positioning, navigation, and timing (PNT) services in European Union (Figure 8), with a focus on enhancing the use of European GNSS (Galileo and EGNOS) across multiple sectors. The plan emphasises the importance of resilient and reliable PNT services for critical infrastructure, societal needs, and economic activities, while also addressing

vulnerabilities such as GNSS signal interference and the need for backup PNT solutions.

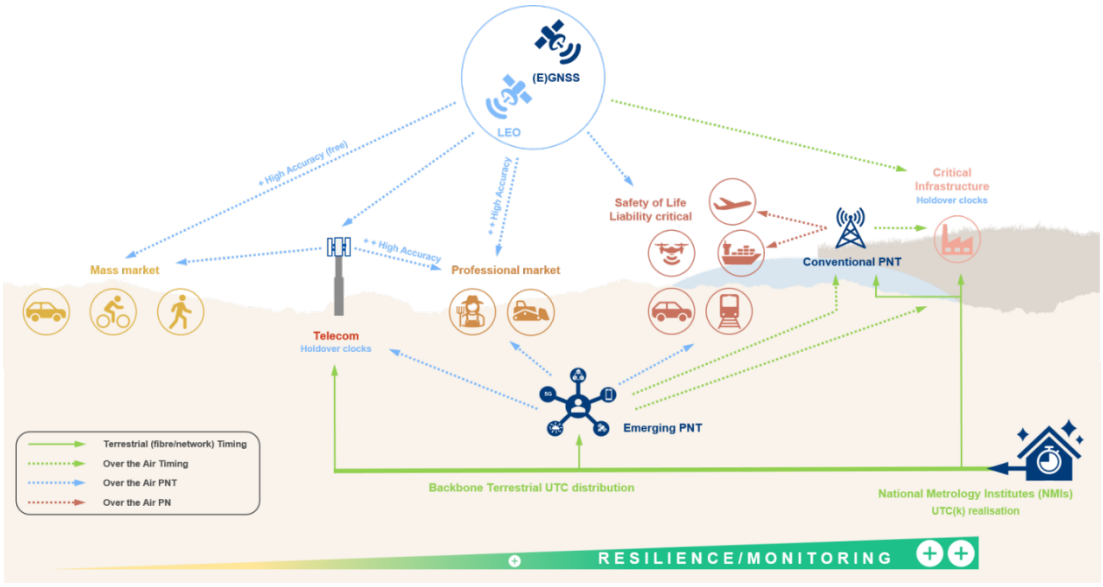


Figure 9 - Vision for the EU PNT

(2) The CNS Evolution Plan, which focuses on the future development of CNS systems within European aviation, is closely aligned with the objectives of the ERNP. The CNS Evolution Plan acknowledges that navigation, as part of the CNS framework, is largely reliant on PNT systems that are outlined in the ERNP. The two plans share common goals, such as enhancing the resilience and reliability of aviation systems through advanced PNT services, especially as aviation moves towards increased digitalisation, automation, and integration of unmanned systems.

(3) To address the increasing vulnerability of GNSS-dependent services, it is essential to consider backup solutions that provide Position, Navigation, and Timing (PNT) independently of GNSS. These technologies are collectively referred to as Alternative PNT (A-PNT). Like GNSS, A-PNT technologies are designed to offer the same critical capabilities.

(4) The range of A-PNT technologies is diverse, providing solutions for time synchronisation as well as full PNT capabilities. The A-PNT test campaign conducted by the Joint Research Centre (JRC) in 2021-22 underscored the critical reliance on GNSS and the need for robust alternatives. In this context, the concept of Complementary PNT (C-PNT) emerges as a critical component. C-PNT refers to technologies that work alongside GNSS to enhance PNT resilience, availability, and continuity.

(5) Both the ERNP and the CNS Evolution Plan will consider A-PNT and C-PNT frameworks to ensure resilient and reliable PNT services across the aviation.

3 Key objectives and challenges of CNS evolution

(1) As part of the preparation of this edition of the CNS Evolution Plan, the CNS Programme Manager initiated a survey with its contributors to verify the existence of issues with the current CNS infrastructure and its evolution, and the need to address them through the development of the CNS Evolution Plan to achieve the vision and strategic goals developed in the previous chapter.

(2) While a full analysis of the problem is beyond the scope of this edition of the Evolution Plan, this chapter aims to provide an initial overview of the challenges identified, their causes and the objectives derived from them.

(3) It also aims to draw an attention of the stakeholder community to short term challenges that CNS PM identified as part of this work. These challenges should be addressed by the relevant actors as soon as possible as they can have a significant impact on the CNS modernisation. These issues are flagged in this chapter.

(4) The 2026 edition of the CNS Evolution Plan will include a more comprehensive and in-depth analysis of the underlying causes of the issues, supported as far as reasonably practicable by tangible and quantified evidence, and using the methodology defined in **Appendix 1**.

3.1 Resilience: Safety, Security and Defence



Strengthening CNS system resilience by implementing robust cybersecurity protocols and backup systems, with the goal of mitigating cyber threats and ensuring secure airspace and safe flight operations.

Why?	As the aviation industry continues to evolve through the advancement of CNS systems, the challenges related to safety, security and defence are becoming more complex. The reliance on interconnected, digital systems exposes aviation infrastructure to new and emerging threats, ranging from cybersecurity risks to physical vulnerabilities of critical systems. Moreover, the integration of civil and military operations in controlled airspace adds another layer of complexity, requiring robust, interoperable, and resilient CNS systems. GNSS services are particularly vulnerable to jamming and spoofing, which can lead to unavailability of service and manipulation of navigation data. GNSS Radio Frequency Interference (RFI) events generate temporary or non-recoverable failure or degradation of PNT resulting in a direct impact on airborne and ground systems, pilots and ATCO workload, and therefore, impact on the safety of the flight operation. As GNSS is a critical component of the CNS infrastructure and a key enabler to CNS based operations, its operations must be

	protected. EASA SIB No. 2022-02R3 [9] includes a comprehensive set of measures to be applied per stakeholder. As CNS systems increasingly rely on space-based assets, they become more vulnerable to space weather events and space object re-entry events. These phenomena can disrupt satellite-based CNS services, affecting both civil and military operations. The consequences of climate change and related extreme weather occurrences are also putting to the test the resilience of CNS ground infrastructures. It could therefore have an adverse impact on CNS performance that has to be considered. In addition, civil military coordination, cooperation and collaboration being key elements to ensure security and defence, it is important to guarantee civil military interoperability and data sharing. In this respect, the new airspace users such as with drones and High Airspace Operations need to be addressed as well.

How?	Proposed actions	Who?
1*	Implement EASA SIB No. 2022-02R3 recommendations to address GNSS outage and alterations leading to CNS degradation	States, Operational Stakeholders and Industry, EASA (for the maintenance of the SIB)
2*	Define local CNS infrastructure plans considering the European MON principles & design criteria	States, Operational Stakeholders in cooperation with CNS PM
3*	Evaluate intervention strategies in relation to ILS CAT I	EASA
4	Investigate dual use of Military CNS systems while fulfilling military requirements for security.	CNS PM, EASA, EDA
5	Ensure civil military data sharing in particular for new entrants.	CNS PM, EDA
6	Ensure Civil Military coordination for GNSS RFI military exercises.	CNS PM, EDA

**** Key CNS Challenge 2025: Address GNSS outage and alterations leading to CNS Degradation***

(1) Since February 2022, there has been an increase in jamming and/or spoofing of GNSS. EASA has analysed recent data from the Network of Analysts and open sources and has concluded that GNSS jamming and/or spoofing has shown further increase in the severity of its impact, as well as an overall growth of intensity and sophistication of these events.

(2) To address the identified issues, States, Operational Stakeholders and Industry should implement recommended mitigating measures contained in EASA SIB No. 2022-02R3.

(3) In particular, States and Operational Stakeholders should ensure that contingency plans, developed in accordance with Regulation (EU) 2017/373 for maintaining operations when CNS systems are out of service, properly address

the risk of GNSS outage and alteration, and ensure that backup systems and procedures are in place to minimise disruptions.

(4) CNS PM's work on the CNS Minimum Operational Network (see key challenge below) will support States and Operational Stakeholders in the identification of non-GNSS based CNS infrastructure able to safely support reversion scenarios and contingency procedures.

(5) The CNS PM will monitor the situation and, in coordination with EASA, identify any additional action that may be necessary to support the development of prevention and mitigation measures which would fall within its remit.

**** Key CNS Challenge 2025: Increase the resilience of CNS (e.g. to GNSS RFI), including application of the European MON principles and design criteria.***

(1) The CNS PM MON Concept and Design Criteria (D01-01) [10] defines the concept of an EU civil CNS Minimum Operational Network (MON) that improves the cost-effectiveness of the CNS infrastructure through optimisation, including the decommissioning of communications, navigation and surveillance assets, while providing an adequate level of resilience and maintaining an acceptable level of safety and security. In particular, the document considers the minimum infrastructure required to support the reversion from GNSS-based CNS infrastructure while enabling aircraft to continue operations in a seamless manner.

(2) The amount of alternative infrastructure included in the national MONs will depend on local system performance, constraints and requirements. Ultimately, the sizing of the MON will be subject to a cost-benefit analysis balancing the cost of infrastructure against the cost of air traffic services and flight disruption.

(3) In April 2021, EUROCONTROL, on behalf of the CNS Advisory Group, produced an economic assessment of the evolution options for the CNS infrastructure [2].

(4) This assessment has been updated to take into account new hypotheses and assumptions developed in the CNS Programme Manager's European MON Concept and Design Criteria deliverable [10]. As with the initial assessment carried out in 2021, a top-down European perspective has been applied, assuming full cross-border data and infrastructure sharing. However, instead of theoretical scenarios based on arbitrary decommissioning percentages, the optimisation of the navigation and surveillance infrastructure is now based on the harmonised principles and MON design criteria proposed in [10].

(5) This remains an idealistic network-level scenario, which may not necessarily correspond to the CNS optimisation strategy of individual states, nor adequately address local requirements and constraints.

(6) Local CNS infrastructure plans will be captured through appropriate monitoring mechanisms. Local and regional constraints will be identified, characterised, compared and, where possible, mitigated. These local plans will be consolidated and incorporated into the optimisation element of future editions of the CNS Evolution Plan.

**** Key CNS Challenge 2025: Evaluate intervention strategies in relation to PBN IR ILS CAT I***

(1) Recent widespread or recurrent local GNSS failures have raised questions about the transition to the exclusive use of performance-based navigation procedures.

(2) EASA is further analysing GNSS vulnerabilities and will evaluate intervention strategies, notably changes to Regulation (EU) 2018/1048 (PBN IR).

(3) One of the key challenges in the coming years regarding PBN implementation, as mandated by the PBN IR, is the exclusive use of PBN after June 2030. This means that ILS CAT I will no longer be permitted for normal operations. A major concern is the low level of SBAS (Satellite-Based Augmentation System) fleet equipage, raising doubts about the entire aviation industry's readiness to fully transition from ILS CAT I to LPV CAT I (SBAS) procedures by 2030. Ongoing discussions at the policymaker level are assessing whether Article 5 of the PBN IR should be amended.

3.2 Operational Efficiency



Support initiatives to improve airspace capacity and operational efficiency, leveraging CNS-enabled airspace management and Trajectory-Based Operations (TBO)

Why?	Operational efficiency is a key priority for the aviation industry as it seeks to meet rising demand while minimising costs, fuel consumption, and environmental impact. The CNS Evolution Plan will have to address the gap between projected air traffic growth and current capacity, ensuring that future CNS systems enable the safe, timely, and efficient movement of aircraft. The shift to Trajectory-Based Operations (TBO) and the wider use of PBN will allow for more flexible and efficient routing, reducing delays, optimising flight paths, and increasing airspace capacity. However, these approaches depend heavily on CNS technologies for precise aircraft positioning, communications, and surveillance, which will require updates and enhancements to existing systems. By 2045, machine-to-machine communications will likely replace routine voice communications, enabling more efficient traffic management [4]. This shift requires investment in modern infrastructure that integrates air, ground-based, and space-based CNS systems in an efficient, service-oriented architecture.	
How?	**Proposed actions**	**Who?**
1	Advance the CNS systems to support strategic (airspace policy), pre-strategic (day-to-day allocation of airspace), tactical (real time use of airspace) airspace management.	Operational Stakeholders[2], Industry
2*	Address current Datalink performance issues and take steps towards multilink to enable future applications.	CNS PM, Operational Stakeholders

[2] Operational Stakeholders are defined as “Civil and military airspace users, civil and military air navigation service providers and airport operators which operate in the airspace referred to in Article 1(4) of Commission Implementing Regulation (EU) 2019/123”.

3*	Address over-interrogation in 1030 MHz band.	CNS PM, EASA, Operational Stakeholders
4	Support implementation of a harmonised CNS infrastructure that enables efficient cross-border operations.	CNS PM, EASA, Operational Stakeholders
5	Propose operational incentives for modernisation of CNS systems.	CNS PM

Key CNS Challenge 2025: Addressing current Datalink performance issues and take steps towards multilink to enable future applications.

(1) VHF Datalink Mode 2 (VDL2) in Europe continues to raise significant reliability and resilience concerns among aviation stakeholders. These challenges are compounded by its limited ability to seamlessly support future traffic, including Air Traffic Services (ATS) B2 and Airline Operational Communications (AOC).

(2) Certain ATN-VDL2 network outages observed during 2024, which resulted in regulations and delays, suggest that the first steps of the multilink infrastructure should be considered. These steps could improve CPDLC performance and support ATS-B2 operations, while focus should also be on improving the ground architecture to ensure resilience.

(3) The Network Manager is conducting a study to assess the current performance and capacity of the European air-ground datalink sub-networks, with a focus on VDL2, and to establish guidelines for forecasting future performance and capacity, taking into account the expected off-loading with the use of IRIS satellite communications (SATCOM) as a complementary link, and the evolution of services, including ATS and AOC. A preliminary report is expected by early 2025.

(4) The SDM, the S3JU and the NM, in close cooperation with the relevant industry, are jointly progressing the implementation of the Multilink Implementation Roadmap [11]. Operational use of ATN-B1 and ATS-B2 over SATCOM IRIS have started in January 2024. The establishment of a common approach for the procurement and management of ATS Common DataLink Services (ACDLS), as done for ground-ground communications with NewPENS, may accelerate problem resolutions and the deployment of new services.

Key CNS Challenge 2025: 1030/1090 MHz bands saturation

(1) Over-interrogation of aircraft transponders by civil and military ground-based surveillance Mode S interrogators and ACAS-equipped aircraft can lead to loss of surveillance - a serious safety risk. Loss of surveillance has already occurred in Europe, and NM's monitoring activities show that Mode S reply rates in the core area of Europe routinely exceed the transponder reply capabilities specified in the equipment standards.

(2) As part of its Operational Excellence Programme, the NM is working with operational stakeholders to address saturation in the 1030 and 1090 MHz bands. First results of this activity include EUROCONTROL guidelines on Mode S interrogator configuration, to be published by the end of 2024. Further material is proposed for development.

3.3 Technological Obsolescence



Promote gradual replacement of outdated CNS systems through a structured transition to modern technologies while ensuring continued operational integrity and adaptability to future needs.

Why?	Technological obsolescence is a growing concern for the aviation industry, as it seeks to modernise its CNS systems while maintaining operational safety and efficiency. Current CNS systems, while reliable and proven, are increasingly unable to support the continued growth in airspace and airport usage expected over the coming decades. As the aviation industry transitions to modern CNS systems, there is an inherent risk of disruption or failure during the adoption process. New technologies, while rigorously tested and approved before deployment, introduce unfamiliar protocols, systems integration challenges, and greater alignment with technologies used in other sectors. This alignment increases the risk of vulnerabilities being shared across industries. Therefore, a phased transition is necessary to avoid a sudden shift to new systems that could cause widespread failures. A mix of legacy and new systems will be required to prevent single points of failure in the case of unexpected disruptions to newer technologies. Given the high costs of new equipment, many operators may continue to rely on older, non-state-of-the-art technologies for as long as possible. This creates a fragmented operating environment where some aircraft are equipped with the latest CNS capabilities while others continue to use outdated systems, complicating efforts to achieve seamless and efficient air traffic management.	
How?	**Proposed actions**	**Who?**
1	Ensure interoperability when introducing new CNS technologies to prevent operational disruptions during the transition.	CNS PM, Industry, EASA, EASCG
2	Prioritise a phased and governed evolution from old to new CNS technologies by ensuring that legacy systems remain in place until new systems are fully operational and proven reliable.	CNS PM, EASA, Operational Stakeholders

3.4 Spectrum and Frequency Management



Promote enhanced compliance with global spectrum allocation guidelines and promote updates to legacy CNS systems to improve spectrum efficiency, reduce interference, and ensure resilience in the evolving aviation landscape.

Why?	The radio frequency spectrum is a scarce natural resource with finite capacity limits and for which demand is constantly increasing. In the context, the introduction of 5G technology has already highlighted potential conflicts between non-aeronautical radio spectrum needs and aeronautical stakeholders, but also the potential harmful interference with critical onboard equipment such as radio altimeters. Furthermore, the lack of efficient use of the spectrum dedicated to aviation safety and lack of adaptability of avionics to the evolving spectrum environment will expose aviation CNS services further to interference and, inevitably, to increase in costs to deploy fixes. Further growth in commercial aviation and the increased data demands makes it important to improve and optimise avionics and CNS equipment, incorporating spectrum efficient and compliant RF (Radio Frequency) technologies. This optimisation/modernisation effort will make it possible for CNS systems to evolve to meet the growth of civil air transport as well as to accommodate the introduction of new aviation uses in a more environmentally sustainable manner. Frequency management focuses on the proper utilisation and assignment of specific frequencies within the allocated spectrum. For aviation, ensuring interference-free access to relevant frequency bands is crucial to maintaining the reliability of CNS systems and air-ground interactions. Interference from non-aviation sources, such as mobile telecommunications, can significantly disrupt operations and compromise safety.	
How?	**Proposed actions**	**Who?**
1*	Contribute to an EU Aviation Spectrum Strategy, outlining foreseeable risks and associated mitigating actions for European stakeholders, aimed at assuring efficiency and longevity of future CNS infrastructure.	CNS PM, EASA, States
2	Address spectrum compatibility integrally at an early stage of CNS system development, to lessen adverse impact on future safety and on equipment lifecycle costs.	CNS PM, Industry, States
3	Establish collaborative channels under suitable governance including aviation regulators,	States

	telecommunications regulators, and industry stakeholders to develop and socialise spectrum policies in ITU and ICAO with long term benefits to the European ATM network.	
4	Actively participate to EASA, EUROCAE, ETSI, SESAR, CEPT meetings and collaborate with those entities in order to make sure that aeronautical interests are properly taken into account in European telecommunications regulation and standards.	CNS PM, EASA, Industry, States
5	Foster civil-military spectrum coordination and frequency allocation.	NM, States, Operational Stakeholders

**** Key CNS Challenge 2025: Develop CNS Spectrum Report, contributing to the aviation spectrum strategy***

(1) One of the critical challenges for CNS Evolution in 2024 is to establish a holistic spectrum strategy and its associated policy actions. The goal of the strategy is to assure continued long-term access to suitable spectrum meeting the growing CNS demands, while tackling short-term issues like interference. To address these challenges a robust game plan is needed that lays out the key milestones and how to achieve them. A CNS Spectrum Report will be developed providing actionable recommendations working towards these milestones, and describing concretely what needs to be done in the short term and in the longer term. The policy themes in the spectrum report will set the foundation for a long-term European Spectrum Strategy safeguarding aviation's access to spectrum and allowing European ATM a path to transition to more capable future CNS technologies as the operational need evolves. In working towards these goals, the spectrum report will provide the necessary alignment between a consolidated European and Global vision for aviation spectrum (in ICAO) and the broader EU socio-economic policies. The strategy is therefore pivotal to driving CNS evolution while protecting aviation interests in a rapidly evolving spectrum landscape. The report will be updated biennially to capture the evolving needs, with the first edition planned by the end of 2024.

(2) The first edition of the spectrum report will lay out key objectives such as the deployment of more spectrum-efficient and interference-resilient CNS systems, optimisation of obsolescent infrastructure, and the protection of aviation spectrum for both current and future CNS technologies. It will also consider the broader strategic risks to CNS that have potential for operational disruption, such as interference compatibility with other public services that rely on spectrum like 5G and broadcast services. It will also address coordination with military and non-aviation stakeholders to balance spectrum use without compromising aviation safety. Collaboration with key stakeholders will therefore be essential to socialise these spectrum policies, and to assure clarity and stability to European aviation stakeholders.

3.5 Implementation Challenges



Deliver unified CNS implementation programme that fosters consistent adoption and deployment across European ANSPs and operators, reducing fragmentation and ensuring smooth, coordinated implementation

Why?	The successful implementation of the CNS Evolution Plan faces numerous challenges. These include regulatory and technical hurdles, financial constraints, the integration of new services, and the complexity of ensuring system interoperability across various domains. While new CNS technologies offer the promise of improved operational efficiency, safety, and environmental benefits, their implementation is often hindered by cost, fragmented infrastructure, and stakeholder coordination issues. Certifying new CNS systems, both for ground infrastructure and airborne equipment, is a lengthy and resource-intensive process. Industry standards, such as those developed by EUROCAE, RTCA, and ARINC as well as EUROCONTROL Specifications, play a crucial role in supporting the certification and deployment of CNS technologies. These standards provide a common framework for ensuring the safety, interoperability, and performance of CNS systems across different regions and regulatory environments. The certification process based on the recognised standards and specifications facilitates consistency in system implementations and cross-border operations.	
How?	**Proposed actions**	**Who?**
1	Design the implementation and monitoring process for CNS evolution using existing framework to maximum extend and avoiding any duplication.	CNS PM
2	Develop national CNS evolution implementation plans based on the (European) CNS Evolution Plan, in consultation with civil and military Operational Stakeholders, and share them with CNS PM.	Member States
3	Ensure accurate planning and development of required standards and supporting regulatory materials, in particular certification specifications.	EASA, EASCG
5	Ensure synchronised airborne-ground-space deployment.	CNS PM, SDM, Operational Stakeholders
6	Ensure OEMs early involvement to ensure market readiness.	CNS PM, S3JU, Industry

7	Promote incremental adoption of new CNS technologies, to ensure manageable implementation of CNS for all Airspace Users.	CNS PM, EASA, Operational Stakeholders
8	Prioritise the development of Business Cases for short-term CNS implementation issues.	CNS PM

3.6 Interoperability and Harmonisation



Invest efforts towards achieving broad interoperability of CNS systems across European airspace and at global level

Why?	The current situation of CNS global harmonisation and interoperability faces challenges due to varying levels of technology deployment and regulatory frameworks across ICAO regions. While global initiatives like the ICAO Global Air Navigation Plan (GANP) and the Aviation System Block Upgrades (ASBUs) have made progress toward aligning CNS systems, differences in national priorities and resource availability continue to create gaps in seamless interoperability and harmonisation efforts. Seamless, safe, and efficient global air traffic management will be achieved by driving the interoperability of Communications, Navigation, and Surveillance (CNS) systems across European airspace and aligning with global standards. This will involve overcoming the fragmentation of national standards, synchronising technology rollouts, and ensuring compatibility between legacy and modern systems to enable consistent cross-border operations. Harmonised CNS systems aligned with global standards will support the optimisation of equipment both on the ground and in the aircraft.	
How?	**Proposed actions**	**Who?**
1	Develop harmonised European CNS technology roadmaps through strong collaboration with all relevant national, operational and institutional stakeholders, industry, while aligning with global standards.	CNS PM
2	Contribute to ICAO CNS roadmaps development within ICNSS TF and GANP.	CNS PM
3	Actively contribute to the evolution and development of ICAO Standards and Recommended Practices (SARPs).	CNS PM, EASA, States
4	Foster international collaboration with key global partners (e.g. USA) to synchronise CNS technology rollouts.	CNS PM, EASA

5	Ensure backward compatibility with existing systems while phasing out obsolete technologies based on the proper consultation with the users.	Operational Stakeholders, Industry, EASA

3.7 Human and Social Aspects



Ensure availability of skilled human resources to support CNS modernisation.

Why?	Availability of skilled human resources play a critical role in ensuring the safe, efficient, and reliable operation of CNS systems, this has been recognised a global issue by ICAO. As air traffic continues to grow and new technologies are integrated into the aviation ecosystem, addressing human factors, including the needs of Air Traffic Controllers, Pilots, Air Traffic Safety Electronics Personnel, and other operational staff, is essential to achieving the objectives of the CNS Evolution Plan. The transition to new CNS technologies will likely require significant changes in the training and competency development of operational staff, including Air Traffic Controllers, Pilots and Air Traffic Safety Electronics Personnel. The introduction of new CNS systems will necessitate the development of new training programs for staff. Re-training will be required to ensure that personnel are proficient in operating, maintaining, and managing the new systems. As new CNS systems are developed, greater emphasis must be placed on ensuring that systems are easy to operate and maintain and do not introduce unnecessary complexity for humans that are operating and maintaining them.	
How?	**Proposed actions**	**Who?**
1	Ensure availability of CNS related training for stakeholders addressing the CNS modernisation and optimisation, as well as human factors.	Training institutions
2	Ensuring that CNS systems are designed according to human factors best practices.	Industry, Operational Stakeholders
3	Ensure involvement staff associations in CNS evolution process to identify and resolve potential human factors issues early, reducing risks and improving the acceptance of new technologies.	CNS PM

4	Address personnel competence and training requirements early, reducing risks and facilitating easier implementation of new technologies.	EASA

3.8 Economic and Financial Aspects



Promote alignment of CNS investments with sustainable finance principles and develop financial incentives to support CNS evolution

Why?	The deployment of new CNS infrastructure requires substantial investment in ground-based, space-based and airborne systems, while the benefits of such investments may not be realised until years after the initial deployment. Retrofitting older aircraft with systems is often prohibitively expensive, slowing the pace of implementation. This creates difficulties in securing funding for these projects, as stakeholders may be hesitant to invest in technologies whose short-term benefits are unclear in the absence of appropriate incentives. Without proper investment planning, there is a risk that systems will be deployed prematurely or without sufficient support. This planning must consider the expected lifespans of individual systems, fleet turnover rates, and the need to minimise downtime during transitions. As new CNS technologies are deployed, there is a growing need to optimise existing legacy systems to avoid unnecessary costs.	
How?	**Proposed actions**	**Who?**
1	Develop a stable investment planning reference that ensures the synchronisation of upgrades across the air, ground and space sides of CNS infrastructure	CNS PM, SDM
2	Develop a robust CBA to support CNS modernisation and optimisation	CNS PM, SDM
3	Establish financial incentives or cost-sharing models (e.g. New Entrants) that promote early adoption and equipage of new CNS systems.	CNS PM, PRB, EC, SDM

3.9 Environmental Sustainability



Minimise the environmental footprint of CNS infrastructure

Why?	CNS systems enable more direct flight paths and smoother descents, reducing fuel burn and CO_2 and non-CO_2 emissions. By optimising flight trajectories, CNS systems help reduce unnecessary altitude changes, holding patterns, and extended taxi times, all of which contribute to a lower environmental impact. CNS technologies can also help mitigate the impact on communities near airports by enabling more precise and predictable flight operations that reduce noise pollution and local environmental degradation. The optimisation of CNS infrastructure leads to significant environmental benefits by reducing the need for redundant equipment and minimising the environmental footprint of aviation infrastructure (reduced energy consumption and emissions). Finally, monitoring the environmental impact of the CNS infrastructure and raising an awareness among CNS community can be improved in the future.	
How?	**Proposed actions**	**Who?**
1	Prioritise deployment of CNS technologies supporting greener ATM operations	CNS PM, Operational Stakeholders
2	Develop a set of indicators/parameters to monitor the environmental impact and sustainability of the CNS infrastructure and its evolution	CNS PM, EASA, SDM, PRB
3	Share the best environmentally friendly practices when managing the CNS infrastructure and promote greening of CNS infrastructure.	CNS PM, Operational Stakeholders

4 Overview of CNS Systems

(1) The next step in building a CNS Evolution Plan is to identify critical path for the European CNS infrastructure to address the problems and meet the objectives outlined in the previous chapter.

(2) In anticipation of this phase, the CNS PM has initiated a survey of contributors to identify current and emerging CNS technologies that should be considered in the development of deployment scenarios, their screening, and their assessment against the above objectives.

(3) Therefore, this initial edition of the CNS Evolution Plan provides an inventory of all the identified candidate CNS technologies, whether they are Operational, Ready for Deployment, and under Research and Development. The primary objective is to reach a consensus on the CNS systems and technologies to be included in the scope of the plan. Additionally, the methodology for the CNS Evolution Plan development, outlined in **Appendix 1**, is critical for ensuring a structured and coherent evolution process that will be refined in future iterations.

(4) By focusing on both, systems in operation and those under research and development, this section prepares the groundwork for establishing performance-based frameworks, harmonisation strategies, and identifying areas requiring technological advancement. These details set the stage for the more comprehensive CNS roadmap that will follow in subsequent editions of the CNS Evolution Plan.

(5) Following legend is used to highlight system referred to in different CNS domains. As well as symbols are used to distinguish whether the system is In Operation, Ready for Deployment, and under Research and Development



Figure 10 - Colour code and Symbols legend for CNS Systems overview

(6) It is acknowledged that the current content of this inventory is primarily focused on the CNS requirements associated with certified aircraft and current flight rules. The 2026 edition of the CNS Evolution Plan will enhance coverage of new entrants, including unmanned aircraft, innovative air mobility vehicles, high-altitude platform systems (HAPS), supersonic aircraft, trans-atmospheric and suborbital vehicles, and support the latest concepts for U-space and higher airspace operations. This may include considerations for remote identification, command and control link and electronic conspicuity.

4.1 COM Systems Overview

(1) The Air-Ground Communications inventory included below focuses on subnetworks captured on the dashboard. Other aspects, such as the transition

from ATN/OSI to IPS network-layer protocols, will need to be considered as well in building the CNS roadmaps for the 2026 edition.

4.1.1 A-G Communications Systems in Operation

(1) These are currently operational Air-Ground Communications Systems (Figure 11) that facilitate communication between aircraft and ATS Unit(s) via ground stations across various phases of flight, ensuring seamless coordination and operational safety in global airspace. These systems include:

a) **High Frequency Data Link (HFDL)**: Provides long-range data communications over oceanic and remote areas.

b) **High Frequency (HF) Voice Communications**: Supports voice communications in oceanic and remote areas where VHF is unavailable.

c) **Satellite Communications (SATCOM)**: including legacy and advanced versions such as Iridium Certus, VIASAT Classic AERO H+, IRIS, and SB-S, which is a series of satellite-based systems used for both voice and data communications, providing global coverage, particularly in areas where ground-based systems are limited.

d) **VHF Data Link Mode 0/A (VDL Mode 0/A)**: An early data communications system used in continental airspace.

e) **VHF Data Link Mode 2 (VDL Mode 2)**: A widely deployed system for digital communications in the VHF band, supporting AOC and Air Traffic Services.

f) **Very High Frequency (VHF) Voice Communications**: The primary voice communications system for air traffic control in most continental airspace.

(2) Each system serves a unique operational purpose, offering a mix of voice and data capabilities that contribute to comprehensive air-ground communications.

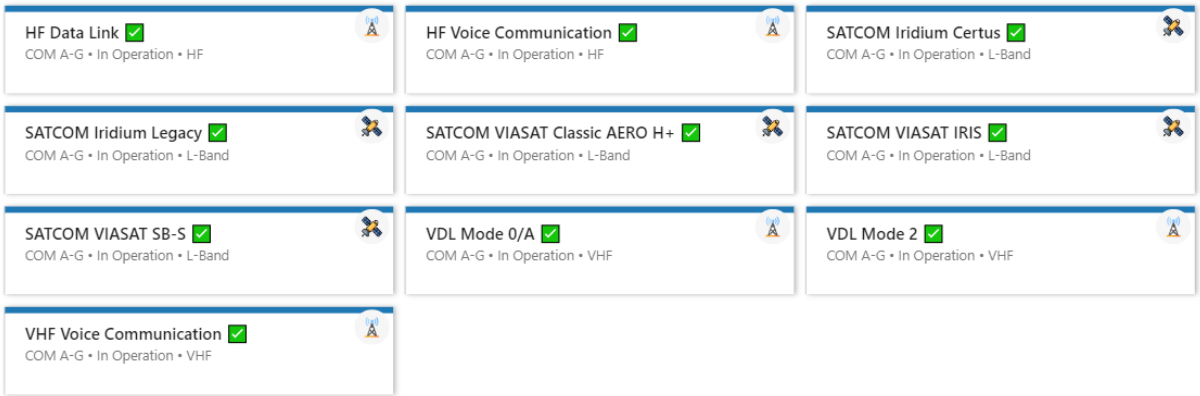


Figure 11 - A-G Communications Systems in Operation

4.1.2 A-G Communications Systems under Research and Development

(1) Air-Ground Communications Systems under Research and Development (**Figure 12**) are emerging technologies aimed at enhancing the future of air traffic management (ATM). These systems are being developed to address increasing demands for data capacity, resilience, and global coverage in communications systems. Key systems under research and development include:

a) **Hyperconnected Air Traffic Management (HCATM)**: HCATM is the capability to use non-safety broadband links to complete safety spectrum

links, to support safety applications (i.e. ATC datalink). This system leverages non-protected aeronautical radio spectrum, which offers flexibility to use existing telecommunication infrastructure but also requires safety link as a back-up.

b) **L-band Digital Aeronautical Communications System (LDACS) -** is a technology in development to provide robust, high-speed, and secure digital data communications in the L-band frequency, offering improvements in bandwidth and latency over current systems like VHF.

c) **Satellite-based Very High Frequency (VHF) Communications (SB-VHF)**: This system aims to extend the reach of traditional VHF communications via satellite and available airborne VHF communications capabilities, enabling voice and data communications in remote and oceanic regions where ground infrastructure is limited or unavailable.

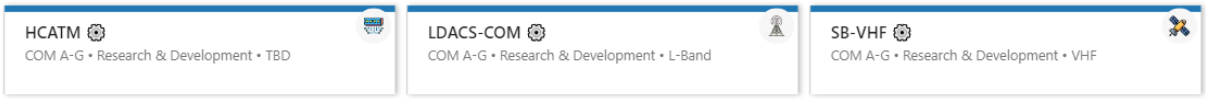


Figure 12 - A-G Communications Systems under Research and Development

4.1.3 A-G Communications Systems Ready for Deployment

(1) From the Ground-ground communications perspectives **Aeronautical Mobile Airport Communications System (AeroMACS) Datalink** is considered as 'Ready for Deployment'. Airborne equipment prototype was developed but require certification and ground equipment is already deployed for ground-ground communications.

(2) AeroMACS facilitates secure and efficient communications for airport surface operations.

4.1.4 G-G Communications Systems in Operation

(1) Operational Ground-Ground Communications Systems (**Figure 13**) facilitate essential communications between air traffic management (ATM) facilities. These systems encompass a wide range of voice communications interfaces and ensure the smooth exchange of information and coordination between air traffic control centres, airports, and other aviation stakeholders. The key systems currently in operation include:

a) **Aeronautical Fixed Telecommunication Network (AFTN)**: Facilitates the transmission of aeronautical information and messages (flight plans, weather updates, Notices to Airmen, etc.), between airports, air traffic control units, airlines, meteorological services, and other aviation stakeholders.

b) **Aeronautical Message Handling System (AMHS)**: An enhanced messaging system that replaces AFTN, offering greater data capacity and more secure communications capabilities.

c) **Voice Communications System (VCS) Voice over Internet Protocol (ATM VoIP)**: Enables digital voice communications between air traffic service units with improved quality and reliability.

d) **Voice Communications System (VCS) Multi-Frequency Code (ATS-R2)**: Supports simultaneous voice communications enhancing coordination capabilities.

e) **Plain Old Telephone System (POTS)**: Traditional telephony system still in use for voice communications in certain operational scenarios.

f) **Proprietary Voice Communications**: Custom voice communications systems used for specific operational needs.

g) **Public Subscriber Telephone Network (PSTN)**: Publicly available telephone system used for general communications purposes within ATM operations.

h) **Voice Communications System (VCS) Air Traffic Services Q Signal (ATS-QSIG)**: A system used to establish direct and reliable voice communications between air traffic service units using coded signals.

i) **Voice Communications System (VCS) Direct Speech**: Supports immediate voice communications between air traffic controllers, critical for real-time decision-making in air traffic management.

(2) These systems form the backbone of ground-ground communications in air traffic management, ensuring reliable, secure, and efficient coordination necessary for safe aviation operations.

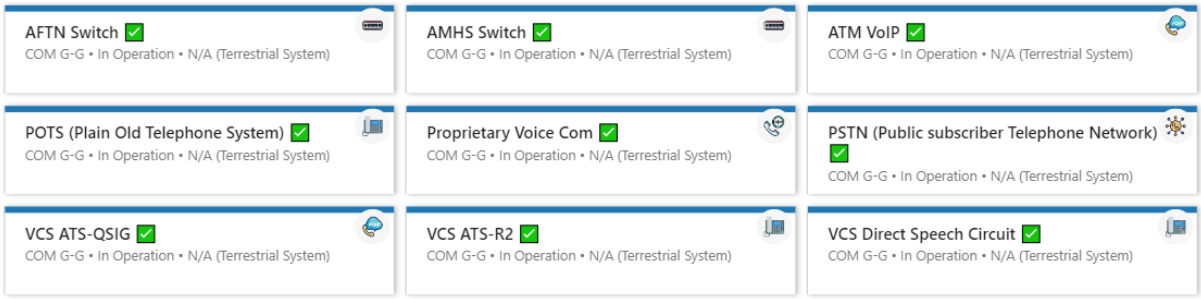


Figure 13 - G-G Communications Systems in Operation

4.1.5 G-G Communications Systems Ready for Deployment

(1) From the Ground-ground communications perspectives **System-Wide Information Management (SWIM) Technical Infrastructure (TI) Yellow Profile** is considered as 'Ready for Deployment'. It facilitates the secure and interoperable exchange of data across the aviation community, enhancing collaboration and situational awareness.

(2) SWIM TI Yellow Profile is part of the DS-GE.CER/DEC — Issue 1 for the Conformity assessment of ATM/ANS equipment, which is referred to in AMC & GM associated with Regulation (EU) 2023/1768 (see appendix). Therefore, SWIM TI Yellow Profile is subject to declaration in accordance with Regulation (EU) 2023/1768 and 2023/1769.

4.2 NAV Systems Overview

4.2.1 Navigation Systems in Operation

(1) The chapter on Operational Navigation Systems provides an overview of the various systems currently in use (Figure 14). These systems range from satellite-based technologies to traditional ground-based navigation aids. The following systems are described:

a) **Global Navigation Satellite System (GNSS) – Single frequency (SF) GPS**: US-operated satellite-based navigation system (satellite constellation) providing position for navigation through single-frequency signals. GNSS

constellation signals in civil aviation must always be used together with aircraft-, space- or ground-based augmentation systems.

b) **Global Navigation Satellite System (GNSS) - Receiver Autonomous Integrity Monitoring (RAIM)**: Aircraft-based augmentation system to the SF GPS constellation providing integrity monitoring for navigation.

c) **Global Navigation Satellite System (GNSS) - Aircraft Autonomous Integrity Monitoring (AAIM):** Aircraft-based augmentation system to the based on the integration of GNSS signals and additional on-board navigation sensors, providing position accuracy and integrity monitoring for navigation.

d) **RNAV system**: A navigation system which permits aircraft operation on any desired flight path within the coverage of ground or space-based navigation aids or within the limits of the capability of self-contained aids, or a combination of these. An RNAV system may be included as part of an FMS.

e) **RNP system**: navigation system which permits aircraft operation on any desired flight path within the coverage of station-referenced navigation aids or within the limits of the capability of self-contained aids, or a combination of these. An RNP system requires on-board performance monitoring and alerting. An RNP system may be included as part of an FMS.

f) **Advanced Required Navigation Performance (A-RNP) FMS function**: Supports highly accurate flight path tracking, enabling efficient route management in performance-based navigation (PBN) environments.

g) **Barometric Altimeter**: Altimetry system that measures altitude using air pressure.

h) **Barometric Vertical Navigation (Baro-VNAV)**: Aircraft-based guidance system that supports vertical navigation for descent and approach phases.

i) **Distance Measurement Equipment (DME)**: Air-and-Ground based system that provides the slant range distance from an aircraft to a ground station, used for en-route and terminal navigation as well as approach procedures. Commonly collocated and associated to an en-route (VOR) or approach (VOR, ILS) landing aids.

j) **ILS/DME**: a variant of DME used to provide distance to the runway during ILS precision approach operations. Use for en-route and terminal operations cannot be assumed universal and is dependent on station configuration and avionic architecture.

k) **Enhanced Flight Vision Systems (EFVS)**: Aircraft-based system that improves pilot visibility in low-visibility conditions, enhancing situational awareness during landing.

l) **GNSS - Single frequency, single constellation (SFSC) Ground-Based Augmentation System (GBAS) GAST C**: Ground-based augmentation system to a single GNSS constellation providing GNSS corrections and integrity monitoring for navigation only on one frequency. It supports precision approach and landing for Category I (and, in some cases, Category II). It can support arrival, departure and non-precision approaches through a GBAS positioning service.

m) **Instrument Landing System (ILS)**: Ground-based system that provides lateral and vertical guidance for aircraft during approach and landing, supporting precision landings in all weather conditions. Defined in three categories in increasing degree of performance and safety criticality (Categories I, II and III).

n) **Inertial Navigation System / Inertial Reference System (INS/IRS)**: Aircraft-based system that provides autonomous navigation information, such as position and velocity. Can be further differentiated in independent/isolated INS which functions, without external references and hybrid INS, commonly found on newer aircraft types, which is regularly disciplined by other sensors.

o) **Non-Directional Beacon (NDB)**: A low to medium wave ground-based navigation aid that in combination with ADF (see below) provides 'bearing-to' information to aircraft.

p) **Automatic Direction Finder (ADF)**: Aircraft-based system that receives NDB signals. Guides aircraft toward or away from NDB ground stations.

q) **Precision Approach Radar (PAR)**: Ground-based system that assists air traffic controllers in guiding aircraft for precision approaches through air-to-ground voice communications, particularly in military operations.

r) **Radio Altimeter**: Aircraft-based altimetry system that measures the aircraft height above the ground, crucial for landing and low-level flight operations. Operational range up to 8 000 feet.

s) **GNSS – Single frequency, single constellation (SFSC) Space-Based Augmentation System (SBAS)**: Space-based augmentation system to a single GNSS constellation providing corrections and integrity monitoring for navigation only on one frequency. Supports enroute, terminal, non-precision approach (NPA) and, in some cases, also approach/landing for APV and/or Category I procedures.

t) **Tactical Air Navigation System (TACAN)**: Ground-based system that provides bearing and distance information to military aircraft, similar to VOR/DME for civilian aircraft. The ranging signal can be configured readable to ICAO-compliant civilian DME interrogators which then perceive it as they would a ground based DME transponder. Commonly collocated and associated to an en-route or approach VOR navigation aids. The bearing signal is readable to a TACAN receiver only.

u) **Very High Frequency Omnidirectional Range (VOR)**: A ground-based system providing radial to/from guidance to aircraft, used for en-route, terminal and non-precision instrument approach navigation. Commonly collocated and associated with a DME in a VOR/DME station.

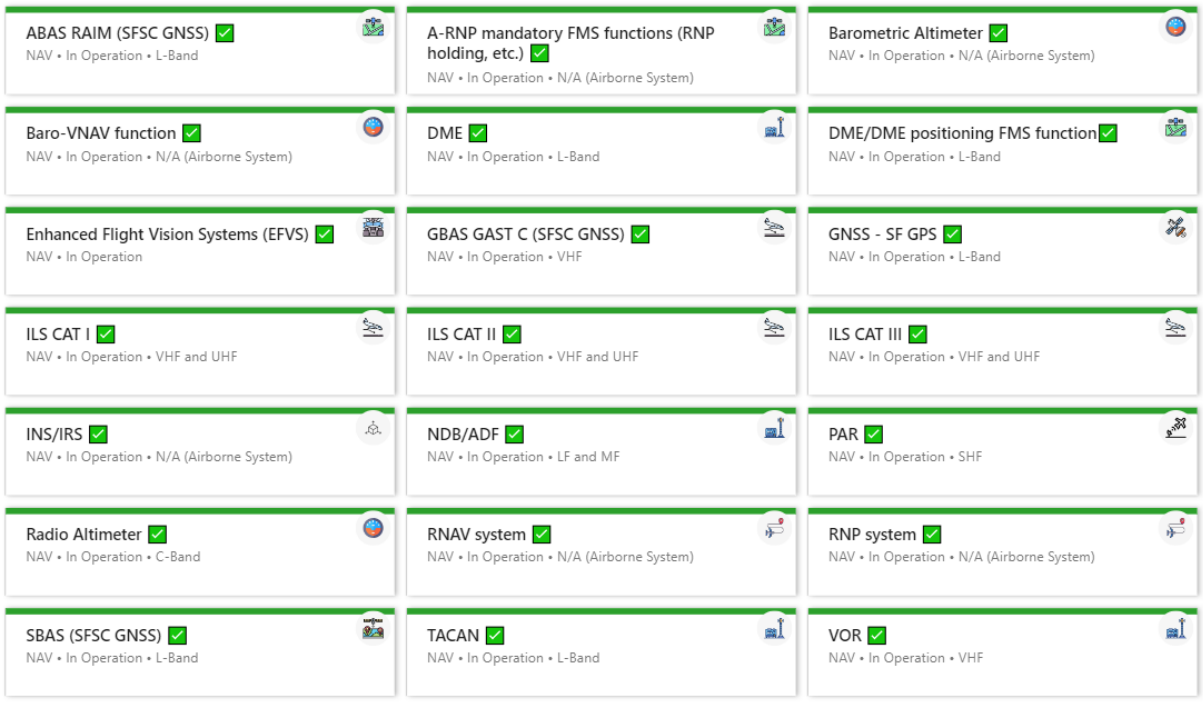


Figure 14 - Navigation Systems in Operation

4.2.2 Navigation Systems under Research and Development

(1) These systems represent the next generation of navigation aids and are designed to meet the growing demands of modern airspace operations. Below is a summary of the key systems under Research and Development (Figure 15):

a) **Global Navigation Satellite System (GNSS) – Dual Frequency (DF) GPS**: US DOD-operated satellite-based navigation system (satellite constellation) providing position for navigation through dual-frequency signals. GNSS constellation signals in civil aviation must always be used together with aircraft-, space- or ground-based augmentation systems.

b) **Global Navigation Satellite System (GNSS) - Dual Frequency (DF) Galileo**: EU-operated satellite-based navigation system (satellite constellation) providing position for navigation through dual-frequency signals. GNSS constellation signals in civil aviation must always be used together with aircraft-, space- or ground-based augmentation systems.

c) **Global Navigation Satellite System (GNSS) - Horizontal-Advanced Receiver Autonomous Integrity Monitoring (H-ARAIM)**: An enhanced aircraft-based augmentation system to the GPS and Galileo constellations providing integrity monitoring to ensure precise horizontal positioning for navigation.

d) **Global Navigation Satellite System (GNSS) - Vertical-Advanced Receiver Autonomous Integrity Monitoring (V-ARAIM)**: An enhanced aircraft-based augmentation system to the GPS and Galileo constellations providing integrity monitoring to ensure precise horizontal and vertical positioning for navigation.

e) **Enhanced Distance Measurement Equipment (eDME)**: An upgraded DME system designed to improve accuracy and reliability in providing slant-range distance for navigation, as well as allowing to reduce DME interrogations while increasing accuracy and reliability.

f) **GNSS - Dual Frequency Multiple Constellation (DFMC) Ground-Based Augmentation System (GBAS) GAST E**: Ground-based augmentation system to several GNSS constellations providing GNSS corrections and integrity monitoring for navigation on two frequencies. It can support precision approach and landing for Categories I, II and III. Additionally, in some cases it can support arrival, departure and non-precision approaches through a GBAS positioning service.

g) **L-band Digital Aeronautical Communications System - Navigation (LDACS-NAV)**: Ground-based communications system that integrates communications and navigation functions, utilising the L-band frequency to enhance data transmission for navigation purposes.

h) **Low-Earth Orbit Positioning, Navigation, and Timing (LEO-PNT) System**: A satellite-based positioning system in low-Earth orbit to provide robust navigation and timing services, offering an alternative to traditional GNSS.

i) **Mode N**: A next-generation ground-based navigation system designed to offer improved positioning capabilities with enhanced security.

j) **Multi Distance Measurement Equipment (DME) Positioning FMS Function**: Combines data from multiple DME stations to enhance positioning accuracy in flight management systems (FMS).

k) **Modular Alternative Positioning, Navigation, and Timing (A-PNT) FMS Function**: A flexible navigation solution designed to provide alternative positioning and timing services when GNSS is unavailable.

l) **Modular Complementary Positioning, Navigation, and Timing (C-PNT) FMS Function**: Enhances navigation resilience by providing complementary PNT solutions alongside GNSS.

m) **GNSS - Dual Frequency Multiple Constellation (DFMC) Satellite-Based Augmentation System (DFMC SBAS)**: Space-based augmentation system to several GNSS constellations providing corrections and integrity monitoring for navigation in dual frequency. Supports enroute, terminal, non-precision approach (NPA) and, in some cases, also approach/landing for APV and/or Category I procedures.

n) **Terrain Aided Navigation** represent new onboard navigation solution based on Terrain map, Radar altimeter, Barometer and Inertial sensors aiming at low visibility conditions and GNSS denied or degraded environment for En route phase.

o) **Radar Based Navigation** represent new onboard navigation solution based on radar aiming at low visibility conditions and GNSS denied or degraded environment for approaches phase.

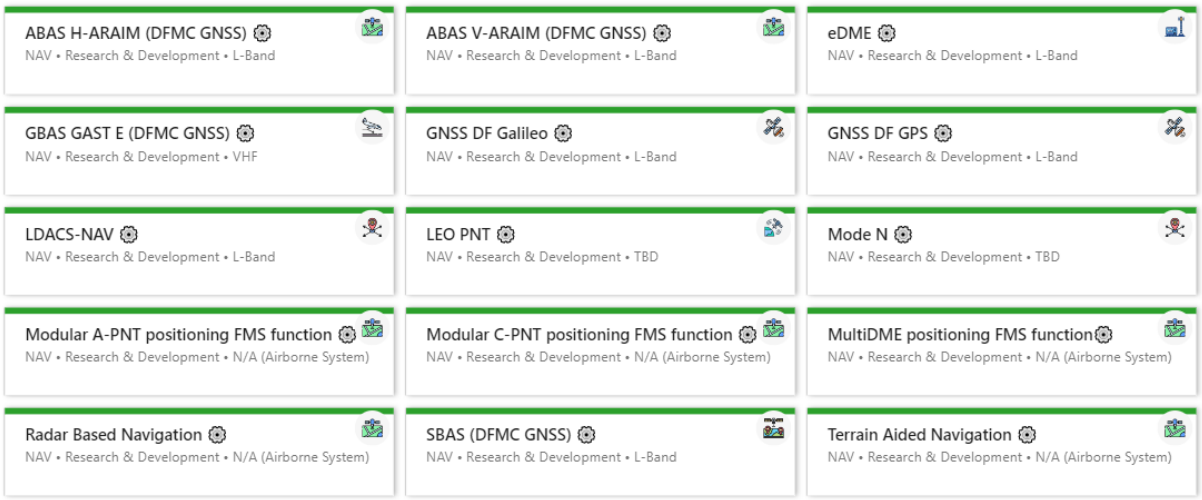


Figure 15 - Navigation Systems under Research and Development

4.2.3 Navigation Systems Ready for Deployment

(1) The following system has reached maturity level TRL6, and is therefore considered as 'Ready for Deployment', even though no certified ground equipment exists today:

a) **GNSS - Single Frequency, Single Constellation (SFSC) Ground-Based Augmentation System (GBAS) GAST D.** It is a Ground-based augmentation system to a single GNSS constellation providing GNSS corrections and integrity monitoring for navigation only on one frequency. It can support precision approach and landing for Categories I, II and III. Additionally, in some cases it can support arrival, departure and non-precision approaches through a GBAS positioning service.

4.3 SUR Systems Overview

4.3.1 Surveillance Systems in Operation

(1) This is an overview of the Surveillance Systems currently in operation (Figure 16), which are critical for maintaining situational awareness, ensuring aircraft separation, and enhancing safety in airspace operations. These systems provide real-time monitoring of aircraft positions and movements, both in the air and on the ground. The key operational systems include:

a) **Automatic Dependent Surveillance-Broadcast OUT (ADS-B OUT)**: Broadcasts an aircraft's position and other data to air traffic control and other equipped aircraft, enhancing surveillance and traffic monitoring.

b) **Satellite-based Automatic Dependent Surveillance-Broadcast OUT**: Extends ADS-B coverage to remote and oceanic regions using satellite technology, ensuring global surveillance capabilities.

c) **Automatic Dependent Surveillance-Broadcast IN (ADS-B IN)**: Enables aircraft to receive real-time position information from other aircraft equipped with ADS-B OUT, improving situational awareness for pilots as well as enabling new airborne applications enhancing safety and efficiency etc.

d) **Airport Multilateration (MLAT)**: Utilises multiple ground-based receivers to determine the position of an aircraft by measuring the time difference of signal arrival.

e) **Surface Movement Radar (SMR)**: Monitors aircraft and vehicle movements on the airport surface, improving safety and efficiency in ground operations.

f) **Primary Surveillance Radar (PSR)**: Detects aircraft without the need for onboard equipment, providing non-cooperative surveillance by reflecting radar signals off the aircraft. It can also provide weather information.

g) **Primary Surveillance Radar (PSR-3D) with 3D target determination**: Primary surveillance radar system that provides three-dimensional positioning (height, range, and azimuth) of targets with integrated weather monitoring capabilities.

h) **Surveillance Data Distribution**: Ensures the reliable distribution of surveillance data to relevant air traffic management (ATM) systems for presentation, decision-making and coordination.

i) **Surveillance Data Processing (Tracker)**: Integrates surveillance data from various sources to provide accurate and continuous aircraft position tracking.

j) **Secondary Surveillance Radar (SSR) Mode A/C**: A radar system that interrogates aircraft transponders to receive basic identification and altitude information.

k) **Secondary Surveillance Radar (SSR) Mode S**: An enhanced version of SSR, providing more detailed aircraft information, reducing interference, and improving overall surveillance capability and performance.

l) **Traffic Alert and Collision Avoidance System II (TCAS) version 7.1**: An onboard collision avoidance system that monitors nearby aircraft to prevent mid-air collisions by issuing traffic and resolution advisories.

m) **Traffic Alert and Collision Avoidance System II version 7.1 with Hybrid Surveillance**: Enhances TCAS II version 7.1 by integrating additional surveillance inputs, reducing interrogation rates.

n) **Wide Area Multilateration (WAM)**: Provides surveillance by combining signals received from a network of ground stations over a wide area. A scaled up MLAT, as the name suggests.

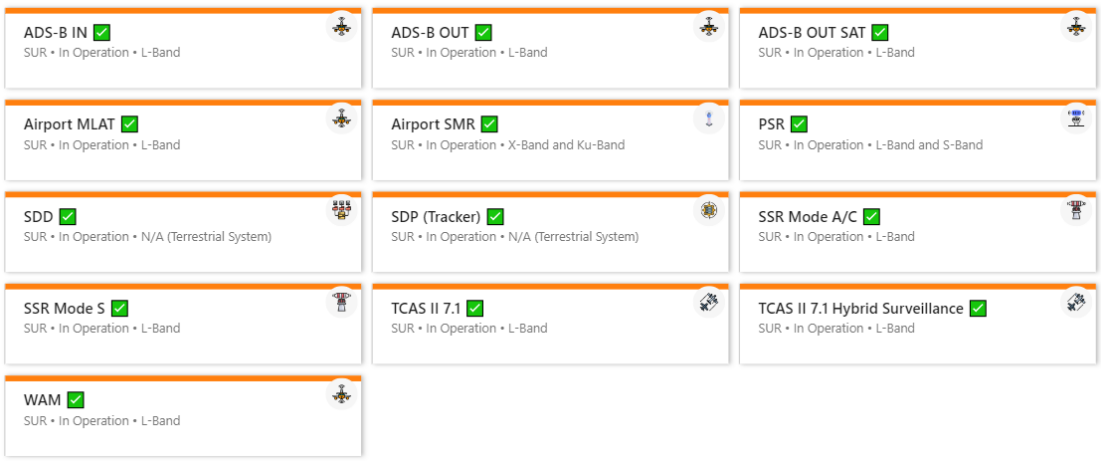


Figure 16 - Surveillance Systems in Operation

4.3.2 Surveillance Systems under Research and Development

(1) This is an overview of the advanced surveillance systems currently under Research and Development, aimed at enhancing the safety, coverage, and adaptability of future air traffic management systems. These systems focus on improving collision avoidance, surveillance of unmanned aircraft, and extending surveillance capabilities in areas with limited infrastructure. The key systems under Research and Development include:

a) **Airborne Collision Avoidance System Passive (ACAS Xp)**: A collision avoidance system that operates without emitting active signals, relying on passive surveillance to detect and avoid nearby aircraft, reducing interference with other systems.

b) **Airborne Collision Avoidance System for Rotorcraft (ACAS Xr)**: Specifically designed for helicopters, this system provides enhanced collision avoidance capabilities in complex and low-altitude environments.

c) **Airborne Collision Avoidance System for UAS and Small UAS (ACAS Xu/sXu)**: Developed for unmanned aircraft systems (UAS) and small drones, this system enables safe integration into airspace by providing real-time traffic awareness and collision avoidance.

d) **Non-Cooperative Surveillance Systems - Multi/Bi-Static Primary Surveillance Radar (NCS MSPSR/BSPSR)**: A next-generation radar system that improves the detection of non-cooperative targets (aircraft without transponders) using multiple radar sources, offering enhanced coverage and accuracy.

e) **Optimised Downlink Airborne Parameter (DAP) and Downlink Airborne Parameter Server**: Improves the efficiency and reliability of surveillance by enabling optimised real-time transmission of important airborne parameters to ground systems.

f) **Satellite-based Wide Area Multilateration (WAM SAT)**: Expands the coverage of traditional multilateration systems by utilising satellite technology to track aircraft over a wider area, particularly in remote and oceanic regions, aiming at increasing resilience over continental areas, while enabling optimisation in the proximity of small airports.

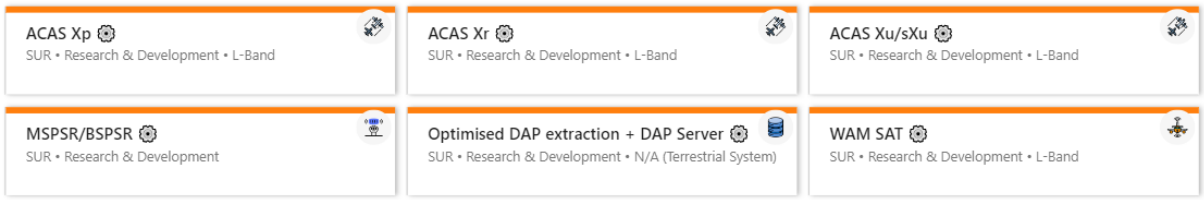


Figure 17 - Surveillance Systems under Research and Development

4.3.3 Surveillance Systems Ready for Deployment

(1) The following system is considered as 'Ready for Deployment' in Surveillance:

a) **Airborne Collision Avoidance System for Active Surveillance / Operation Specific (ACAS Xa/Xo)**: An onboard collision avoidance system that uses a probabilistic model to calculate collision risks instead of relying on fixed thresholds like TCAS. ACAS Xo is an extension to ACAS Xa designed for particular operations, like closely spaced parallel approaches or DNA (Designated No Alert).

(2) EASA published the Opinion No 03/2024 for the introduction of ACAS Xa in the single European sky (SES). Although ACAS Xo is not mandated in the SES, the draft ETSO-C219a for ACAS Xa/Xo referring to EUROCAE ED-256 is part of NPA 2023-04, which is currently under process at EASA.

5 CNS Roadmap and Strategy

This is a placeholder for the content of the 2026 edition of the CNS Evolution Plan.

5.1 Scope

5.2 CNS Strategy

5.3 CNS Implementation Roadmap

5.4 Key Milestones and Challenges

5.5 Synchronisation of Ground, Space and Airborne Deployments

5.6 Critical Path Definition Implementation and Performance Considerations

5.7 Frequency and Spectrum Management

5.8 Optimisation Opportunities and Future MON

5.9 Human Dimension Aspects

5.10 Business Case Analysis

5.11 Risk Management

6 Implementation and Monitoring

(1) The 2026 edition of the CNS Evolution Plan will include a more comprehensive and in-depth analysis of the underlying causes of the issues, as well as a refined assessment of the options; supported as far as reasonably practicable by tangible and quantified evidence and using the approved methodology.

(2) The plan will encompass all Member States within the Single European Sky (SES) as well as the broader European Civil Aviation Conference (ECAC) Area. The long-term perspective will extend up to 2050, aligning with technological advancements and evolving operational requirements to ensure that CNS systems can meet the increasing demand for air traffic management, sustainability, and global competitiveness.

(3) Commonly agreed actions part of the approved Evolution Plan will be proposed as Implementation Objectives and associated stakeholder lines of actions. Implementation will be monitored using the established reporting mechanisms on implementation progress through the "LSSIP+", the Local Single Sky ImPlementation Tool.

(4) The plan will also consolidate the CNS modernisation aspects with the optimisation of legacy CNS infrastructure, taking into account the national MONs designed using the common design principles but based on local requirements and constraints, and captured through the same LSSIP+ tool and the associated CNS questionnaire.

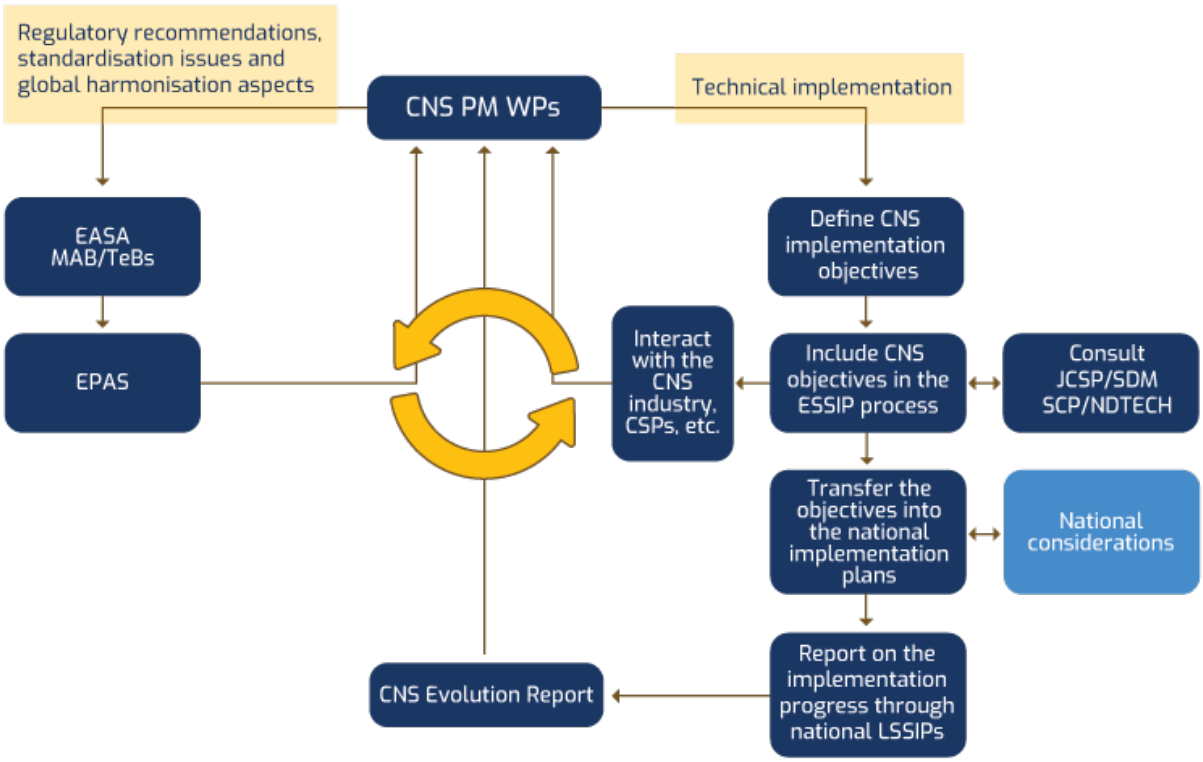


Figure 18 - Proposed CNS Evolution Plan monitoring and reporting process

Appendices

1 Development Methodology of Evolution Plan

1.1 Introduction

(1) This appendix describes the proposed approach for the development of the CNS Evolution Plan. It is based on best practice in impact assessment and is consistent with ICAO's performance-based approach.

(2) Given the reduced scope of this edition of the CNS Evolution Plan, the methodology has not been fully exercised and validated. It will be reviewed and refined as it is applied throughout the lifecycle of the CNS Programme Manager and in particular in the development of the 2026 edition of the CNS Evolution Plan, which will include a full CNS roadmap.

(3) The approach is freely based on impact assessment best practices.

An impact assessment is a process comprising a structured analysis of problems and corresponding responses. It develops objectives and alternative options and assesses their impacts.

(4) The approach is also broadly in line with ICAO's performance-based approach (Table 1):

Key components of an impact assessment [12]	**ICAO performance-based approach [13]**
1. Consultation of interested parties	1. Define/review scope, context and general ambitions/expectations
2. Definition of the problem	2. Identify opportunities, issues and set (new) objectives
3. Definition of the objectives.	3. Quantify objectives
4. Strategic options	4. Select solutions to exploit opportunities and resolve issues
5. Analysis of the impact per se	5. Implement solutions
6. Comparison of the options	6. Assess achievement of objectives
7. Follow-up and assessment	

Table 1- Performance-based approach and impact assessment comparison

(5) A graphical summary of the process is represented on Figure 19, showing links between CNS Evolution Plan methodology steps and document content.

(6) In recognition of the complexity of the process and long times involved in the rollout, a robust risk management regime will be applied across all stages of development and deployment of the roadmap.

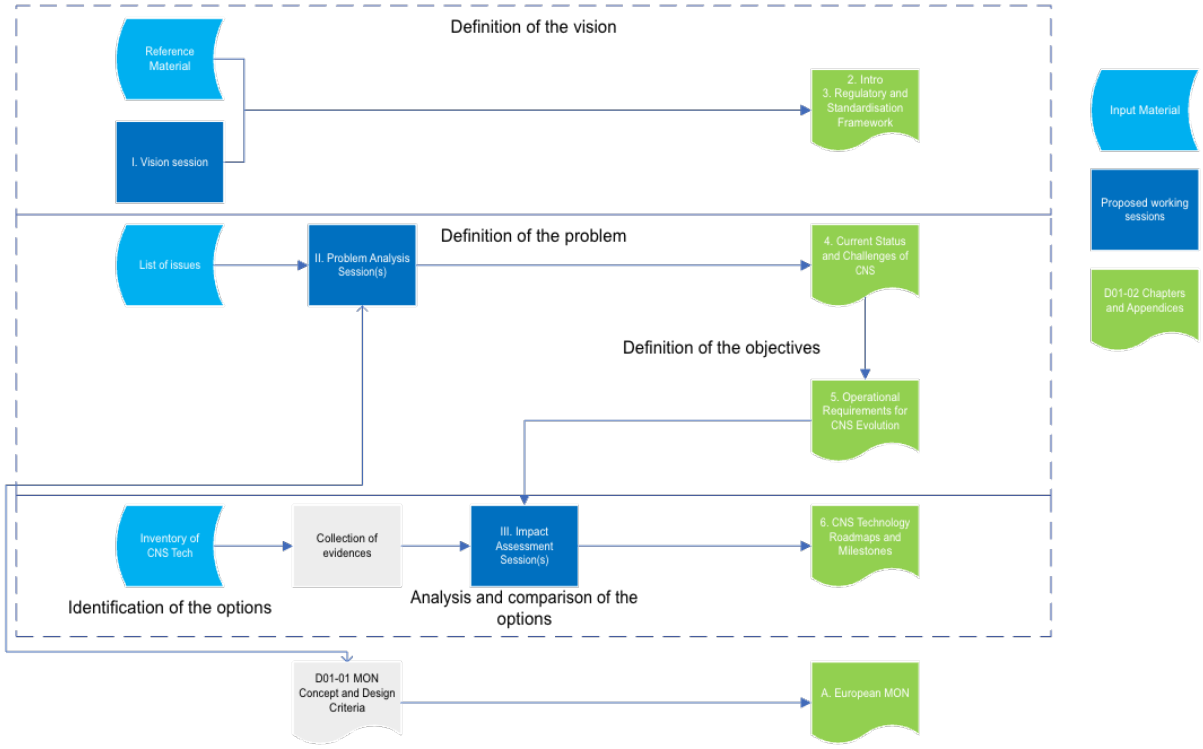


Figure 19 - CNS Evolution Plan development process overview

1.2 Contribution from and consultation of interested parties

(1) The CNS Evolution Plan, led by the Network Manager (NM) in its role as CNS PM, is the result of collaborative efforts across operational stakeholders, states, and other interested parties.

(2) The core CNS PM team will be supported by the following resources during the **elaboration** of the CNS Evolution Plan:

(3) CNS experts from EUROCONTROL's Network Manager Directorate and Civil-Military Coordination CNS units.

(4) A dedicated team of CNS experts from the SESAR Deployment Manager (SDM).

(5) CNS experts nominated after a call for interest to the Joint CNS Stakeholder Consultation Platform (JCSP), representing a wide range of operational stakeholders, states, the industry as well as international federations of staff associations.

(6) Focal points of CNS PM's institutional stakeholders (EASA, EDA, S3JU, PRB, EC).

(7) The draft CNS Evolution Plan will subsequently be subject to a wide **consultation** through both the JCSP and SDM's own consultation platform. Comments will be collected and addressed individually, in a Comment-Response Document accompanying the updated CNS Evolution Plan.

(8) **Approval** will happen through the Network Cooperative Decision Making (CDM) working arrangements, i.e. through the Network Directors of Technology Group (NDTECH) and the Network Management Board (NMB).

(9) For further details, see CNS PM's Programme Management Plan [14].

1.3 CNS Evolution Plan Development Steps

1.3.1 Definition of the vision

(1) It is proposed to develop a vision in the introductory chapter of the CNS Evolution Plan.

(2) The vision paints a picture of the future. It answers the question, "Where do we want to go?" A well-crafted vision describes the desired state or outcome.

(3) It is proposed to dedicate one of the first CNS Evolution Plan working sessions to develop such a vision for the CNS infrastructure (at 2030, 2040 and 2050 time horizons).

(4) The CNS Advisory Group report [1] and the European ATM Master Plan [4] provide elements that could be reused in this exercise; however, the resulting vision should remain concise and impactful. Ideally, it should be a few sentences or a short paragraph.

(5) The CNS Evolution Plan will outline a strategy how to achieve this vision. It will answer the question, "How will we get there?".

1.3.2 Definition of the problems

(1) The first step is to verify the existence of problems with the current CNS infrastructure and its evolution, and a need to address them through the development of the CNS Evolution Plan to achieve that above vision, i.e. the desired end state. The problem analysis is a crucial step as only a correct diagnosis of the problem and its causes can lead us to the appropriate response. If the problem is ill-defined, it is unlikely that the next stages would identify relevant objectives or effective options.

(2) The objective of the problem definition phase is to answer the following questions:

(3) What are the issues or problems which we are trying to solve through the CNS Evolution Plan?

(4) What is the scale of the problems? Are they relevant or not? Are the problems local or global?

(5) What are the root causes underlying the issues or problems?

(6) Who is concerned by the problems, how and to what extent?

(7) How may the problems develop if nothing is done in addition to the actions already taken or planned by operational stakeholders, states or other parties involved?

(8) The CNS Advisory Group report (in particular its recommendations #3 to #7) and the CNS PM MON Concept and Design Criteria document [10] provide elements that could be reused to feed the problem analysis and develop the problem statement; however, it is important to avoid "backward engineering" where the problem analysis is performed with a specific outcome in mind.

(9) First, a survey has been initiated with contributors to collect arguments and problem perceptions that will need to be further analysed. Problems should far as possible be supported by tangible evidence (ideally information collected based on verifiable and reliable methods by trustworthy and neutral sources).

Stakeholder views are often reflecting interests of certain groups of stakeholders but can complement objective data by giving an indication of the relative importance of the problems.

(10) It is then proposed to dedicate at least one CNS Evolution Plan working session to the problem analysis.

(11) In the case of CNS, it is expected that problems and drivers will be numerous, complex or interrelated. Wherever possible, the problem analysis should try to disentangle complex problems into several simpler ones that perhaps can be addressed separately and use visual aids (such as problem trees) to describe them and to link them through to the (later to be defined) objectives and options.

(12) After completing the problem analysis, and prior to set objectives and consider options, the exercise should be summarised through a succinct **problem statement** that will contextualise the problems and explain why they matter.

1.3.3 Definition of the objectives

(1) Objectives link the problems and their drivers to the options. Setting objectives helps to set the level of ambition and fix the yardsticks for comparing options.

(2) Objectives can be general (e.g. “Improve the cost efficiency of CNS systems”), or specific and setting concretely what the CNS Evolution Plan is meant to achieve. For each problem identified above, there should be at least one, but most likely a set of specific objectives.

(3) When setting objectives, notably specific objectives, it can be useful to reflect on the S.M.A.R.T criteria: objectives should be Specific, Measurable, Achievable, Relevant and Time-bound (i.e. ‘S.M.A.R.T’). Still, they should be broad enough to allow consideration of all relevant alternatives without prejudging a particular solution.

1.3.4 Identification of options

(1) Identifying alternative options will be an iterative process. The aim is to consider as many realistic alternatives as possible and then narrow them down to the most relevant ones for further analysis.

(2) The following four steps are suggested to identify a realistic set of options:

a) Construct a baseline from which the impacts of the options will be assessed.

b) Start by compiling a wide range of alternative options.

c) Identify the most viable options; explain the discarded options.

d) Describe in reasonable detail the key aspects of the retained options to allow an in-depth analysis of the associated impacts.

1.3.4.1 Baseline

(1) The baseline is the benchmark against which other options are assessed. Rather than a “do nothing” scenario, this would more correctly be described as a scenario, without a coordinated CNS Evolution Plan to evolve the CNS infrastructure.

1.3.4.2 Compilation of options

(1) As many options as practical should be considered (in addition to the baseline), thinking outside of the box and allowing all stakeholders' preferred options to be considered, as long as options are closely linked to the drivers of the problems, the problems themselves and the identified objectives of the CNS Evolution Plan.

(2) In anticipation of this phase, a survey has been initiated with contributors to identify current and upcoming CNS technologies. This will need to be complemented by a collection of corresponding evidence (e.g. CBA data from SESAR research project, maturity level, available standards…).

1.3.4.3 Screening of options

(1) Time and resources available being finite, it is important to focus the analysis on viable options only. Excluding options at this stage should be clearly justified. Reasons should be as clear, self-evident and indisputable as possible.

(2) For instance:

a) Technological solutions readiness (including standardisation) may not allow for the implementation of theoretical options within the envisaged time horizon.

b) There might be a total absence of data and other evidence that would allow the further analysis of the option.

c) There is no point in retaining options that do not address the objectives as identified in the problem definition.

(3) Figure 20 shows the illustration of a possible screening process.

(4) For the sake of transparency, an explanation should be given for each candidate option being discarded at this stage. Those discarded options should be saved for further analysis outside of the critical path of the work, or future assessment when maturity progresses, or additional evidence is available.

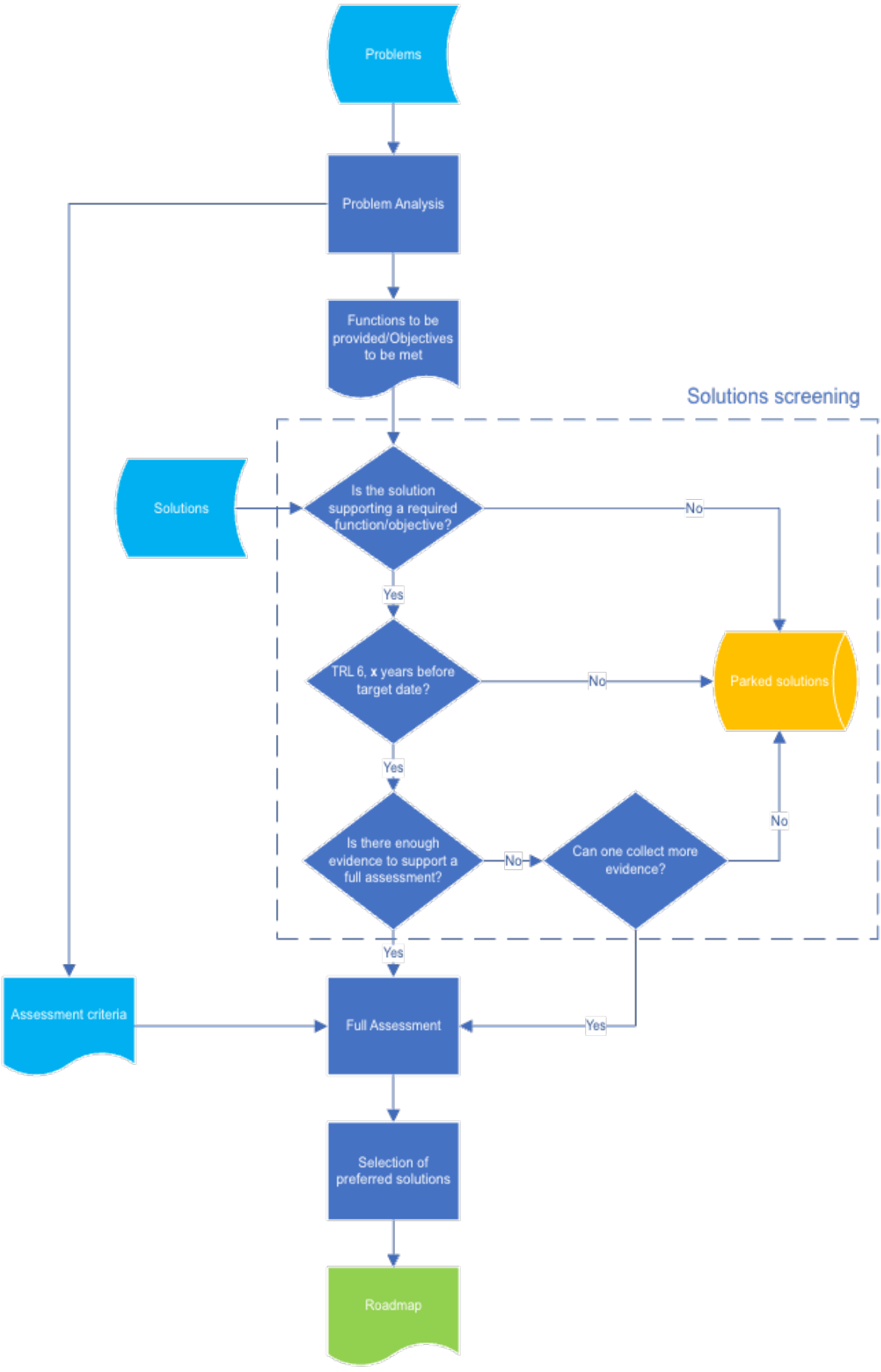


Figure 20 - Solution screening process

1.3.4.4 Description of retained options

(1) Options should be sufficiently well developed to allow to differentiate them based on their ability to meet the identified objectives.

(2) The retained options (including the baseline option) should therefore be described in detail, as alternative CNS Evolution Plans, including technological choices, deployment scenarios, and transition aspects.

1.3.5 Analysis of the options

(1) Each of the options shortlisted will have to be assessed (in comparison to the baseline scenario) against the above-mentioned objectives and according to the following criteria:

a) What is the likely economic impact? What are the costs and benefits induced compared to the baseline scenarios? Are the costs and benefits equally shared between stakeholders or are some parties exposed to disproportionate costs, without any hope for return on investment?

b) What is the likely environmental impact of each the options, how are they expected to contribute to meeting Europe 2050 climate and environmental objectives?

c) Will they have a positive or negative social impact, be it direct or indirect?

d) What will be their impact on safety?

e) What will be their impact on security?

f) What will be their impact on ATM/CNS availability and resilience?

g) What will be their impact on civil-military coordination?

h) What will be their impact in terms of global interoperability? Are the options in line with ICAO's and other regions' plans?

i) What will be their impact in terms of spectrum use and frequency allocation?

j) What is their strategic alignment with other initiatives and policies?

(2) While the economic assessment may deliver a quantitative result, it is likely that the other analyses may only allow a qualitative assessment, which should be translated in a score.

1.3.6 Comparison of the options

(1) A Multi-Criteria Decision Analysis tool, such as the one below will be used to support a structured comparison of the various options based on the above criteria.

(2) Embedded is Infrastructure Australia[3]'s Multi-Criteria Analysis template (Excel).



Assessment Framework 2021 Mu

[3] Infrastructure Australia has been established by the Infrastructure Australia Act 2008 to advise governments, industry and the community on the investments and reforms needed to deliver better infrastructure for all Australians.

(3) Should there be an agreement concerning the equal importance of each criterion, and should criteria be appropriately differentiated to not bias the analysis, a weighting of criteria might not be necessary.

(4) However, in a complex scenario such as the development of the CNS Evolution Plan, criteria will likely need to be weighted to ensure an alignment with policy and objectives.

(5) Weighting could be set using a combination of:

a) surveys to understand stakeholder preferences;

b) workshops of experts or decision-makers being asked to vote on the importance of criteria using a secret ballot approach to reduce cognitive bias.

(6) It is important to agree with the weighting of the criteria before starting the assessment of options.

(7) The CNS Evolution Plan will provide comprehensive and detailed results of the options analysis, classifies the options on the basis of various assessment criteria, and confirm the added value of the preferred option for the CNS Evolution Plan.

1.4 Follow-up and assessment

(1) The CNS PM has, amongst its allocated tasks, the responsibility to monitor the implementation of the CNS Evolution Plan and coordinate the overall CNS Advisory Group Action Plan [3].

(2) The CNS PM will build on and enhance the existing Local Single Sky implementation monitoring process as required to observe the actual progress of the CNS development in Europe. A technological watch will also be maintained to identify potential new solutions and follow the evolving maturity level of existing ones. Finally, an ongoing review of risks and issues will be conducted.

(3) This may lead in a review of the CNS Evolution Plan in an upcoming edition, such as:

a) Reviewing the problem analysis to take account of the evolving CNS environment.

b) Reviewing the baseline scenario to take account of voluntary and regulatory changes.

c) Updating the selection of options, and their assessment, based on new solutions or updated evidence being available.

(4) Ultimately, as implementation of the CNS Evolution Plan progresses, the CNS PM will be able to conduct an ex-post impact assessment, measure the impact of its action and the effectiveness of the preferred option in meeting the above general and specific objectives.

1.5 Risk Management

(1) The 25-year European Communications, Navigation, and Surveillance (CNS) Evolution plan, by its very nature, faces a broad and complex range of risks which can abruptly or over time lead to unforeseen consequences for not only the CNS Evolution plan but also the overall evolution of European ATM. These risks must be carefully identified, tracked and managed to ensure the successful rollout of

new CNS systems supporting Air Traffic Management (ATM) through to 2050. These risks include but are not limited to technical challenges, such as integrating new technologies with legacy systems, issues leveraging improvements through coordinating efforts across multiple stakeholders, and external factors like regulatory updates and evolving cybersecurity requirements. Given the long-term scope of the plan, evolving economic, social, environmental, and technological conditions may also influence its progress. To address these complexities, and in line with best practice in programme and project management, timely and robust risk management is essential, enabling continuous monitoring, proactive mitigation, and adaptive strategies that align with global aviation standards and future industry needs.

(2) To this end, a **risk management methodology**, based on existing project management best practices, will parallel the assessment and scheduling methodology to systematically identify and assess risks, identify mitigations, and assign responsibility and ensure effective monitoring and follow up activities over the course of the CNS roadmaps development and deployment. This approach will serve to ensure the timely deployment of a resilient and efficient European ATM infrastructure for years to come.

(3) This risk management methodology shall be applicable to the roadmap itself and while its output may be of use to stakeholders, in no way will it replace or supersede existing business, security or safety risk management processes or outcomes carried out by stakeholders in relation to their own projects and deliverables.

2 Regulatory and Standardisation Framework

2.1 ICAO Standards Framework and Global Coordination

(1) In the context of Communications, Navigation, and Surveillance (CNS) systems, ICAO plays a pivotal role in establishing global standards and practices essential for maintaining safety, efficiency, and interoperability in international aviation. The effective functioning of global aviation relies heavily on the robust framework of standards and recommended practices (SARPs) established by ICAO. As the aviation industry faces increasing demands for capacity and integration of new technologies, ICAO's commitment to developing, coordinating, and maintaining these standards is vital for the seamless integration and operation of CNS systems across all regions. This commitment is further emphasised by ICAO Assembly Resolution A41-8, which underscores the importance of enhancing CNS resilience by maximising the integration of all suitable ground, space, and airborne components in a complementary and cooperative manner. This includes improving technological capabilities, civil-military operational procedures, and coordination with radio regulatory authorities to ensure the safe and efficient operation of CNS systems.

(2) SARPs related to CNS are developed through a collaborative process that involves member states, industry stakeholders, and technical experts. These standards are embedded within several key Annexes to the Chicago Convention, most notably Annex 10 (Aeronautical Telecommunications), which is dedicated to CNS systems. Annex 10 is divided into volumes addressing different aspects of CNS, including Radio Navigation Aids, communications procedures, communications systems, Surveillance and Collision Avoidance Systems, and Spectrum utilisation (Figure 21). For the CNS Evolution Plan, it is crucial to consider the ongoing revisions and updates to these SARPs, which are driven by the need to accommodate new technologies. These evolving standards ensure that CNS systems remain interoperable, resilient, and capable of supporting future air traffic demands.

(3) To ensure that SARPs related to CNS systems are effectively implemented, ICAO employs several monitoring and auditing mechanisms. The Universal Safety Oversight Audit Programme (USOAP) plays a key role in assessing how well member states adhere to CNS-related standards. The audit findings provide insights into areas where further support or capacity-building might be needed, particularly in regions with limited infrastructure or technical expertise. Additionally, ICAO's regional air navigation plans provide detailed guidance for the implementation of CNS systems at the regional level. These plans are aligned with the ICAO Global Air Navigation Plan (GANP) and are regularly updated to reflect changes in technology and operational requirements. The coordination of these plans ensures that CNS systems across different regions are interoperable and capable of supporting global aviation operations.

(4) As CNS technologies advance, ICAO is adapting its standards-making processes to accommodate new capabilities while ensuring interoperability and safety. This involves the development of performance-based standards and the adoption of industry standards where applicable. The ICNSS-TF is currently working on streamlining the standardisation framework which is crucial for

keeping pace with technological innovation and maintaining global CNS system integration by ensuring that these systems are capable of withstanding disruptions and continue to provide reliable service across all phases of flight. As the aviation industry looks toward the future, ICAO is actively engaged in addressing the challenges posed by emerging CNS technologies. These developments require continuous updates to SARPs to accommodate new capabilities while ensuring backward compatibility with existing systems.

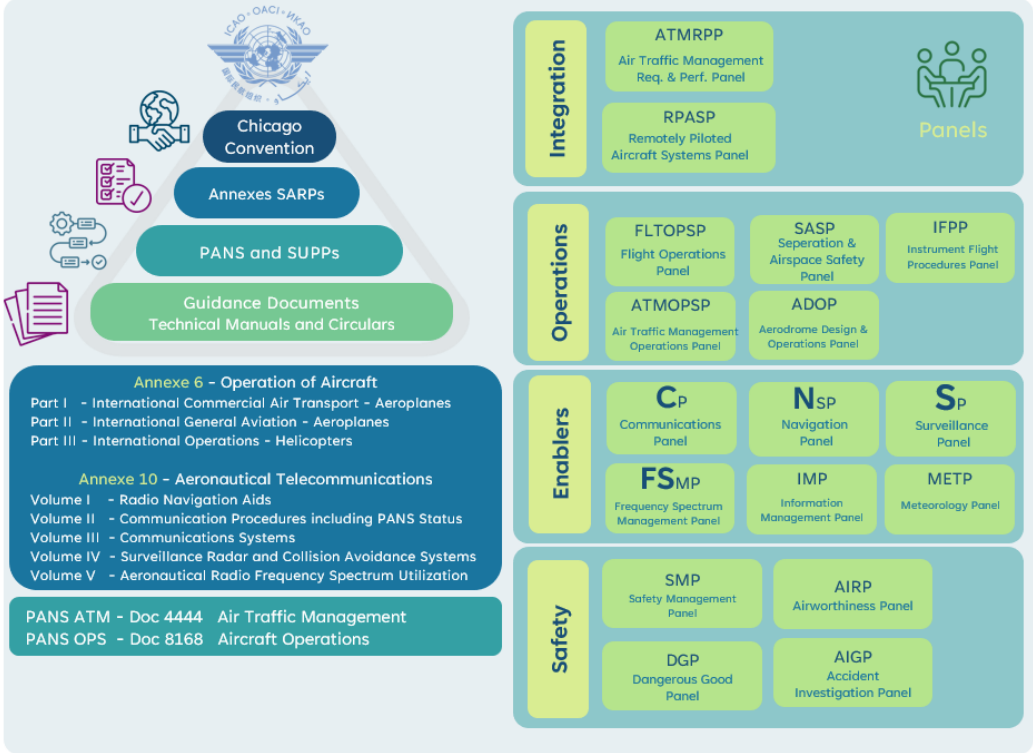


Figure 21 - CNS Framework in ICAO

2.2 EU Regulatory Framework

(1) The key Regulatory Framework elements applicable to CNS are provided at different levels:

a) Regulations:

i) Implementing Regulation (EU) 2023/1770 laying down provisions on aircraft equipment required for the use of the Single European Sky airspace and operating rules related to the use of the Single European Sky airspace.

ii) Regulation (EU) 1332/2011 laying down common airspace usage requirements and operating procedures for airborne collision avoidance.

iii) Implementing Regulation (EU) 923/2012 laying down the common rules of the air and operational provisions regarding services and procedures in air navigation.

iv) Implementing Regulation (EU) 2018/1048 laying down airspace usage requirements and operating procedures concerning performance-based navigation.

v) Implementing Regulation (EU) 2017/373 laying down common requirements for providers of air traffic management/air navigation services and other air traffic management network functions and their oversight.

vi) Regulation (EU) 139/2014 laying down requirements and administrative procedures related to aerodromes pursuant to Regulation (EC) No 216/2008.

vii) Implementing Regulation (EU) 2023/1769 laying down technical requirements and administrative procedures for the approval of organisations involved in the design or production of air traffic management/air navigation services systems and constituents.

viii) Delegated Regulation (EU) 2023/1768 laying down detailed rules for the certification and declaration of air traffic management/air navigation services systems and air traffic management/air navigation services constituents.

b) Acceptable Means of Compliance (AMC) and Guidance Materials (GM):

i) EASA AMC & GM to Implementing Regulation (EU) 2023/1770, Part-AUR.SUR — Issue 1.

ii) EASA AMC & GM to Implementing Regulation (EU) 2023/1770, Part-AUR.COM — Issue 1.

iii) EASA AMC & GM to Implementing Regulation (EU) 2018/1048 on common airspace usage requirements and operating procedures 'AMC & GM to AUR' - Issue 2.

iv) EASA AMC & GM to the Articles of Delegated Regulation (EU) 2023/1768 — Issue 1.

c) Certifications Specifications (CS) and Detailed Specifications (DS):

i) EASA Certification Specifications for Airborne Communications, Navigation and Surveillance (CS-ACNS) - Issue 5.

ii) EASA Certification Specifications for European Technical Standard Orders (CS-ETSO) Amendment 17.

iii) EASA Detailed Specifications and AMC & GM for certification or declaration of design compliance of ATM/ANS ground equipment (DS-GE.CER/DEC) — Issue 1.

iv) EASA Detailed Specifications and AMC & GM for statement of compliance of ATM/ANS ground equipment (DS-GE.SoC) — Issue 1.

(2) An overview of the EU Regulatory Framework applicable to the Communications, Navigation, and Surveillance (CNS) systems is provided in Figure 22.

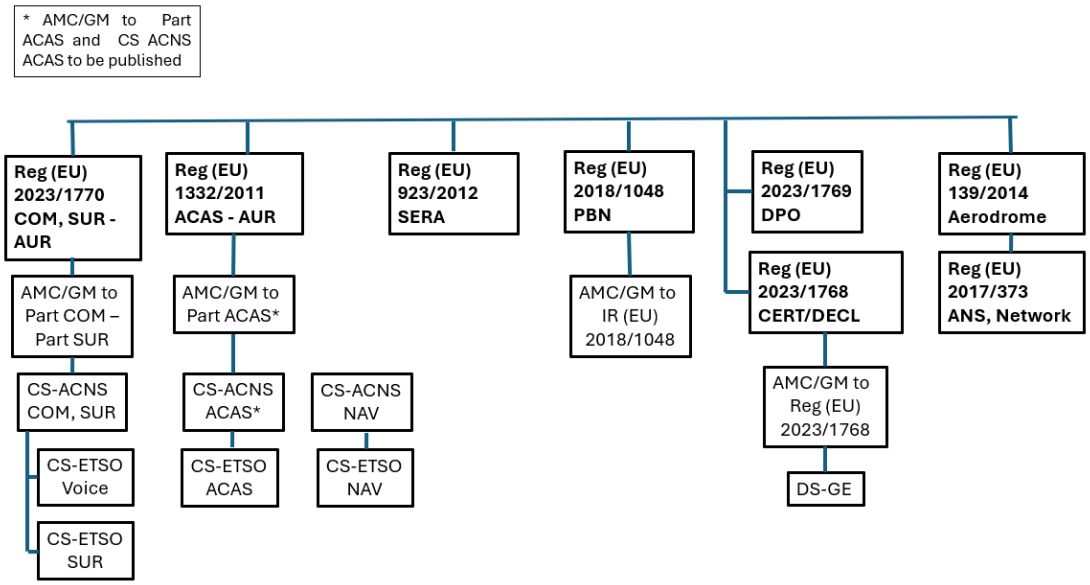


Figure 22 - Overview of the EU Regulatory Framework for CNS Systems

(3) Note: The recital (3) of PBN Implementing Regulation (EU) 2018/1048, which is per se not applicable to airborne navigation systems, mentions the rules referring to the airworthiness certification e.g. SERA (Standardised European Rules of the Air) IR (EU) 923/2012 and Air-OPS (Air Operations) IR (EU) 965/2012). However, there is no AMC to those rules referring to the Certification Specifications and consequently the link between the rules and the CS-ACNS for Navigation is not represented on Figure 22 as for Communications, Surveillance and ACAS.

2.3 European ATM Standards Coordination

(1) Aviation industry standards are critical to ensuring the safety, efficiency and interoperability in Europe's airspace. These standards provide a common framework to guide the modernisation, operation and maintenance of aircraft, air traffic management and communications, navigation and surveillance systems. By complying with these standards, the aviation industry can minimise risks, improve communications and streamline operations, ultimately leading to safer and more reliable air travel.

(2) The European ATM Standardisation Coordination Group (EASCG) plays a key role in this landscape. The EASCG facilitates the harmonisation of ATM standards across Europe and ensures that all stakeholders, including Standard Developing Organisations (SDOs), the aviation industry, air navigation service providers, airlines and regulators, are aligned. This coordination is essential to facilitate the sharing of work among the SDOs thus avoiding the risk of overlapping developments and gaps.

(3) Through its collaborative efforts, the EASCG helps foster innovation and the adoption of new technologies while ensuring compliance with European and global regulations. This not only enhances the safety and efficiency of European airspace, but also contributes to the global aviation community by setting a benchmark for best practice in ATM standardisation.

(4) The EASCG currently coordinates ATM-related standardisation activities essentially stemming from the European ATM Master Plan, in support of Single European Sky implementation.

(5) The main deliverable of the EASCG is the European ATM Standardisation Rolling Development Plan (A-RDP). It brings together all relevant standardisation activities and their status and is updated regularly in order to maintain visibility and awareness of the progress.

(6) It provides a method for the identification and discussion of overlaps, and as a basis for feedback to contributing organisations, to improve overall coordination of standards developments (Figure 23). The process also identifies the technical input from other sources (such as ICAO GANP and ASBUs, EUROCAE, RTCA, …) into the standards plan through the interfaces of the EASCG with the international level.

(7) The A-RDP is instrumental in assessing the maturity of CNS technological solutions and their readiness for deployment and industrialisation. By providing a detailed overview of ongoing and planned standardisation efforts, the RDP helps stakeholders identify the current status and development stages of various CNS technologies. This visibility is essential for determining whether a technology is sufficiently mature to be implemented in real-world operations, and at what time horizon it may be included in the CNS Evolution Plan.

(8) The A-RDP not only reflects current activities but also outlines future standardisation needs. This forward-looking approach helps in anticipating technological trends and preparing for the integration of emerging solutions; the A-RDP will therefore need to address the needs stemming from the approved CNS Evolution Plan.

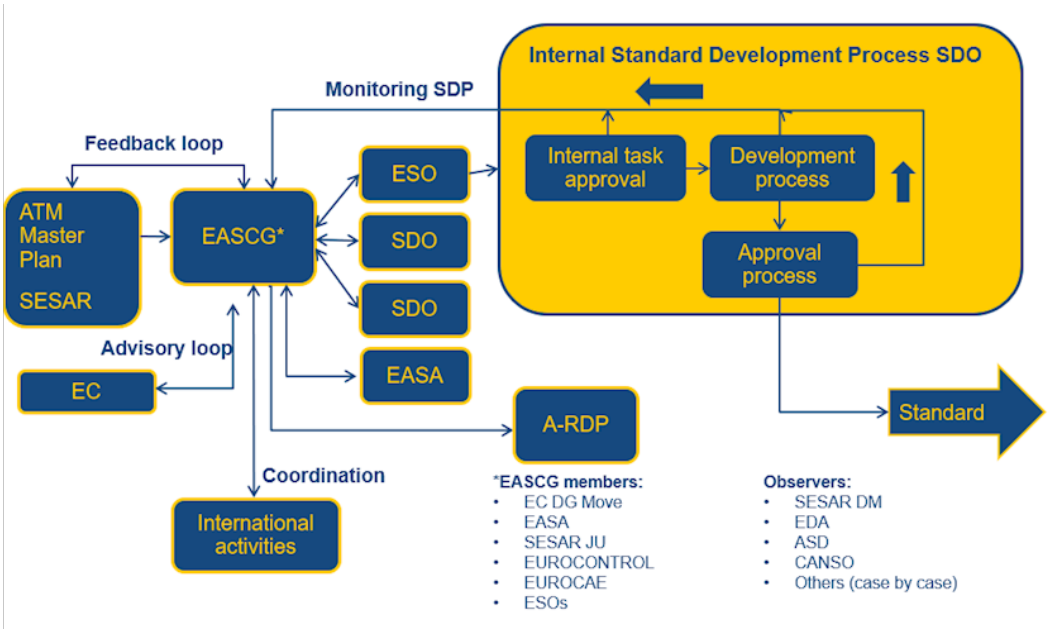


Figure 23 - EASCG Coordination Process

2.4 ITU Radio Regulations Framework

(1) The International Telecommunication Union (ITU) carries out the global management of radio frequency spectrum, which is vital for the functioning of Communications, Navigation, and Surveillance (CNS) systems in aviation. The regulatory framework established by the ITU, particularly through the Radio Regulations (RR), provides the legal foundation for the allocation and use of radio frequencies worldwide, which in turn is transposed to national legislation and enforced in each contracting Member State. The national administration responsible for this is typically the telecommunications regulator.

(2) The RR enacts provisions to protect aeronautical frequencies from interference that could jeopardise safety and disrupt operations by instituting a number of technical and administrative conditions on the assignment and coordination of frequencies, which are again the responsibility of the telecommunications regulator to enforce. Furthermore, the RRs also implement specific measures to ensure that the use of aeronautical frequencies is compliant with the aforesaid technical and administrative measures by requiring CNS equipment to be compliant with the ICAO SARPs. The telecommunications and aviation frameworks are therefore interconnected to assure that aviation's unique spectrum needs are met globally while maintaining avoiding interference with other spectrum users.

(3) The collaboration between ITU and ICAO is thus essential to safeguard the integrity of the radio frequencies that are critical for CNS systems, and to ensure ongoing access to radio spectrum to meet future Air Traffic Management and other aviation needs.

(4) The ITU's World Radiocommunication Conferences (WRCs) are the ITU's instrument for maintaining the RRs. These conferences, held nominally every four years, review and revise the Radio Regulations, and they address spectrum allocation requests from the various sectors including broadcasting, science services, International Mobile Telephony (e.g. 5G), licence free services (e.g. Wi-Fi), government services (such as police, border forces, emergency etc), maritime and aviation services, among others. ICAO actively participates in WRCs as an observing party to represent the radio spectrum interests of aviation, advocating for the protection of aviation spectrum allocations and following decisions that affect CNS systems. The outcomes of these conferences directly influence the availability and integrity of the radio spectrum for the continued safe operation of CNS systems worldwide.

(5) As the representative delegations in the WRC are the telecommunications administrations of each Member State, careful and timely coordination between telecoms and aviation stakeholders is essential. ICAO facilitates this coordination by preparing a (global) Common Aviation Position for each WRC (Figure 24), through its expert body, the Frequency and Spectrum Management Panel which is introduced above. This makes the global and regional coordination between ITU and Aviation Community a cornerstone of global aviation safety.

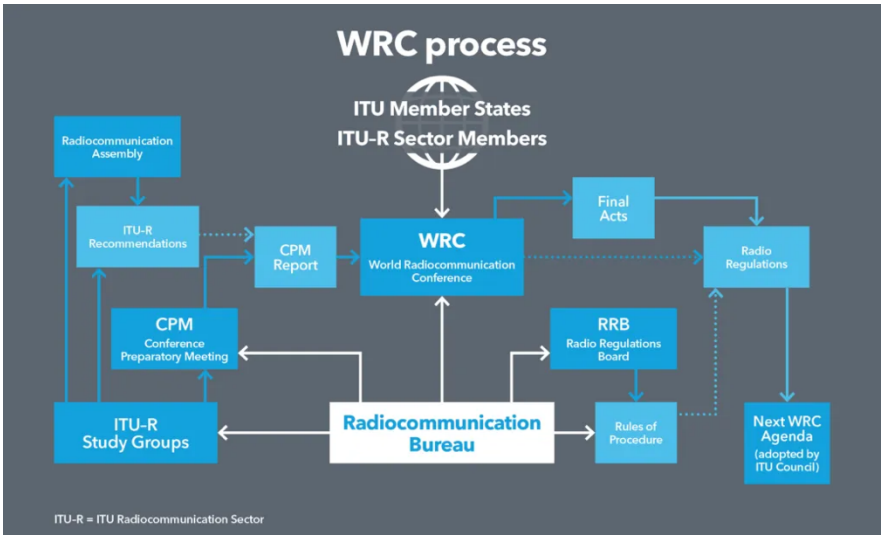


Figure 24 - The World Radiocommunication Conference Process

(6) For the European CNS Evolution Plan, aligning with the ITU regulatory framework is essential to ensure that CNS systems are consistent with global standards and practices. The Evolution Plan must take into account the spectrum allocations and the regulatory resolutions and recommendations established by the ITU, particularly as new technologies and systems are integrated into the European airspace.

(7) The national regulatory authorities are the active participants in ITU processes, particularly WRCs, and will play a key role to advocate for the spectrum needs of CNS systems as well as ensure civil-military coordination for access to the shared spectrum.

3 ATM Master Plan Cross Reference

(1) The ATM Master Plan is the key driver for the entire ATM innovation cycle in Europe and an important input to other strategic European initiatives, including this CNS Evolution Plan.

(2) The European ATM Master Plan 2025 Edition [4] sets out the vision for the Digital European Sky by 2045, the strategic deployment objectives to be implemented by 2035, and the development priorities for the coming years.

(3) This overlaps with the CNS Evolution Plan timeframe which extends up to 2050. The purpose of the present appendix is therefore to provide the traceability between the priorities and objectives set by the ATM Master Plan and the content of this CNS Evolution Plan. In the context of this preliminary 2024 CNS Evolution Plan, it provides links to the candidate CNS technologies identified in Section 4, while it is anticipated that edition 2026 of the CNS Evolution Plan will provide reference to detailed technical recommendations.

3.1 Strategic Deployment Objectives

(1) The strategic deployment objectives (SDOs), in the ATM Master Plan, are prioritised actions covering ATM functionalities that need to be rolled out between 2025 and 2035. These actions are critical for achieving the vision and performance ambitions, and are based on SESAR Solutions, delivered or in the pipeline.

SDO ID	Description
SDO #01	Alerts for reduction of collision risks on taxiways and runways
SDO #02	Optimising airport and TMA environmental footprint
SDO #03	Dynamic airspace configuration
SDO #04	Increased automation support
SDO #05	Transformation to trajectory-based operations (TBO)
SDO #06	Virtualisation of operations
SDO #07	Transition towards high performance of air-ground connectivity (multilink)
SDO #08	Service-oriented delivery model (data-driven and cloud-based)
SDO #09	CNS optimisation, modernisation and resilience
SDO #10	Enable innovative air mobility (IAM) and drone operations

Table 2 - European ATM Master Plan 2025 Strategic Deployment Objectives

(2) Two of the above SDOs are directly related to CNS and include CNS deployment actions: SDO 7 and SDO 9.

(3) **SDO 7 – Transition towards high performance of air-ground connectivity (multilink)**

(1) This strategic deployment objective aims to improve air-ground datalink performance, accelerate technology adoption, and achieve economies of scale while reducing the fragmentation that characterises current datalink implementation.

(2) SDO 7 is composed of 3 different deployment actions which themselves rely on a number of SESAR Solutions. This SDO is directly linked to the modernisation of air-ground datalink which is further studied in the context of the CNS Evolution Plan.

Action #	Deployment Actions	CNS Evolution Plan Content
7.1	Implement future air-ground communications network infrastructure, which supports multilink capability and complete mobility between different datalinks.	Sections: 3.1 CNS Technologies: N/A[4]
7.2	Implement SatCom class B, which enables data and voice communication services using existing satellite technology systems in oceanic, remote, polar, and progressively continental airspace.	Sections: 3.1 CNS Technologies: ☑ SATCOM (VIASAT IRIS)
7.3	Implement VDL-M2 successor (e.g. terrestrial datalink system L-band-digital aeronautical communications system (LDACS), datalink for ATM and AOC operations over commercial communication systems (hyper-connected ATM), and satellite communications for both continental and remote/oceanic regions.	Sections: 3.1, 3.3 CNS Technologies: ⧖ AEROMACS ☼ LDACS-COM ☼ HCATM ☑ SATCOM (Iridium Certus)

Table 3 - CNS Evolution Plan content linked to SDO 7

(3) **SDO 9 - CNS optimisation, modernisation and resilience**

(1) This strategic deployment objective aims to optimise, modernise and increase the resilience of CNS infrastructure in Europe.

(2) SDO 9 is composed of 4 different deployment actions which themselves rely on several SESAR Solutions. The deployment actions exclusively focus on the optimisation of the NAV and SUR infrastructure, even if the overall SDO description is very generic and it could be argued that the whole CNS Evolution Plan is a more detailed view of this SDO, covering the optimisation, modernisation and resilience of Communications, Navigation and Surveillance.

[4] Multilink not being a specific communications technology, it is not currently captured amongst the technological options.

Action #	Deployment Actions	CNS Evolution Plan Content
9.1	Implement a secured surveillance functionality that enables detection and, when possible, mitigation of security threats that could affect the surveillance chain.	Sections: CNS Technologies: N/A
9.2	Implement minimum operational network (MON).	Sections: CNS Technologies: CNS PM D01-01
9.3	Rationalise instrument landing systems (ILS) and implement efficiency measures/methods for more cost-effective maintenance of ILS, providing a link between ICAO Doc. 8071 and national CNS provision.	Sections: CNS Technologies: CNS PM D01-01[5]
9.4	Optimise surveillance, leveraging terrestrial and space-based information.	Sections: CNS Technologies: CNS PM D01-01

Table 4 - CNS Evolution Plan content linked to SDO 9

3.2 Strategic Development Priorities

(1) The European ATM Master Plan 2025 Edition also defines strategic development priorities (DPs), which cover the prioritised actions needed to develop future ATM functionalities and are planned for deployment until 2045. Not all development priorities are necessarily relevant for CNS. IR-02 can be seen as further developing SDO 7 and other IR priorities may benefit from CNS enablers, but it is difficult to assess the CNS relevance of exploratory research priorities at this stage due to their very nature.

[5] The current version of D01-01 does not address ILS optimisation, pending the completion by EASA of the analysis of GNSS vulnerabilities and the necessary changes to Regulation (EU) 2018/1048.

IR	Strategic development priorities	
IR-01	Transformation to trajectory-based operations	Industrial research
IR-02	Transition towards high performance of air–ground connectivity (multilink)	
IR-03	Future en-route and TMA ground platforms	
IR-04	Future airport platform	
IR-05	Autonomy and digital assistants for the flight deck	
IR-06	U3 U-space advanced services, IAM and vertiports	
AR-01	Research to help shape the future regulatory framework for a Digital European Sky	Exploratory research
AR-02	Definition of U4 U-space full services	
AR-03	Integration of the next generation aircraft for zero-/low-emission aviation	
FR-01	ATM impact on climate change	
FR-02	Digital flight rules	
FR-03	Investigate quantum sensing and computing applied to ATM	

Figure 25 - European ATM Master Plan 2025 Strategic Development Priorities

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[12] EU Impact Assessment Handbook.

[13] ICAO Doc 9883, Manual on Global Performance of the Air Navigation System.

[14] CNS PM Programme Management Plan (D00-01).

Abbreviations

Term	Definition
AMC & GM	Acceptable Means of Compliance (AMC) and Guidance Material (GM)
ANSP	Air Navigation Service Provider
A-RDP	ATM Standardisation Rolling Development Plan
ASBU	Aviation System Block Upgrades
ATCO	Air Traffic Control Officer
ATM	Air Traffic Management
ATSEP	Air Traffic Safety Electronics Personnel
CBA	Cost-Benefit Analysis
CDM	Cooperative Decision Making
CNS	Communication Navigation and Surveillance
COM	Communications
CS	Certifications Specifications
CS-ACNS	Certification Specifications for Airborne Communications, Navigation and Surveillance
CS-ETSO	Certification Specifications for European Technical Standard Orders
CSP	Communications Service Provider
DS	Detailed Specifications
EASA	European Union Aviation Safety Agency
EASCG	European ATM Standards Coordination Group
ECAC	European Civil Aviation Conference
EDA	European Defence Agency
EPAS	European Plan for Aviation Safety
ERNP	European Radio Navigation Plan
ESA	European Space Agency
EU	Europe Union
EUSPA	European Union Agency for the Space Programme
GANP	Global Air Navigation Plan
GNSS	Global Navigation Satellite System
IAM	Innovative Air Mobility
ICAO	International Civil Aviation Organisation
ITU	International Telecommunication Union

JCSP	Joint CNS Stakeholder Consultation Platform
MON	Minimum Operational Network
MS	Member States
NAV	Navigation
NDTEC	Network Directors of Technology
NM	Network Manager
NMB	Network Management Board
NSP	Network Strategy Plan
PNT	Positioning, Navigation and Timing
PRB	Performance Review Body
R&D	Research and Development
RR	Radio Regulation
S3JU	SESAR3 Joint Undertaking
SARPs	Standards and Recommended Practices
SDM	SESAR Deployment Manager
SDO	Strategic Deployment Objective
SERA	Standardised European Rules of the Air
SES	Single European Sky
SO	Strategic Objective
SUR	Surveillance
TBO	Trajectory Based Operation
WRC	World Radiocommunication Conference

Table 5 - Abbreviations table

Definitions

Term	Definition
ATM/ANS constituent	means tangible objects such as hardware and intangible objects such as software upon which the interoperability of the EATMN depends (source: Regulation (EU) 2018/1139).
ATM/ANS equipment	means ATM/ANS constituents as defined by Article 3(6) of Regulation (EU) 2018/1139 and ATM/ANS systems as defined by Article 3(7) of that Regulation, excluding airborne constituents, which are subject to Commission Regulation (EU) No 748/2012 (source: Delegated Regulation (EU) 2023/1768).
ATM/ANS system	means the aggregation of airborne and ground-based constituents, as well as space-based equipment, that provides support for air navigation services for all phases of flight (source: Regulation (EU) 2018/1139).
CNS infrastructure	means the aggregation of CNS systems (airborne, ground-based and space-based).
Communications services	means aeronautical fixed and mobile services to enable ground-to-ground, air-to-ground and air-to-air communications for ATC purposes (source: Regulation (EU) No 549/2004).
CNS Dual Use	represents the optimal use of military CNS capabilities to sustain civil requirements.
Navigation services	means those facilities and services that provide aircraft with positioning and timing information (source: Regulation (EU) No 549/2004).
Surveillance services	means those facilities and services used to determine the respective positions of aircraft to allow safe separation (source: Regulation (EU) No 549/2004).
New entrants	include low-level drones, military medium-altitude long-endurance unmanned aircraft systems, automated air taxis, super-high-altitude (FL600+) operating aircraft, next generation supersonic aircraft and electrically propelled aircraft

Table 6 - Definitions table



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